
ASUS ZE500KL

RF Hardware Trouble Shooting Guide

Outline

(*) Debug Tool Required

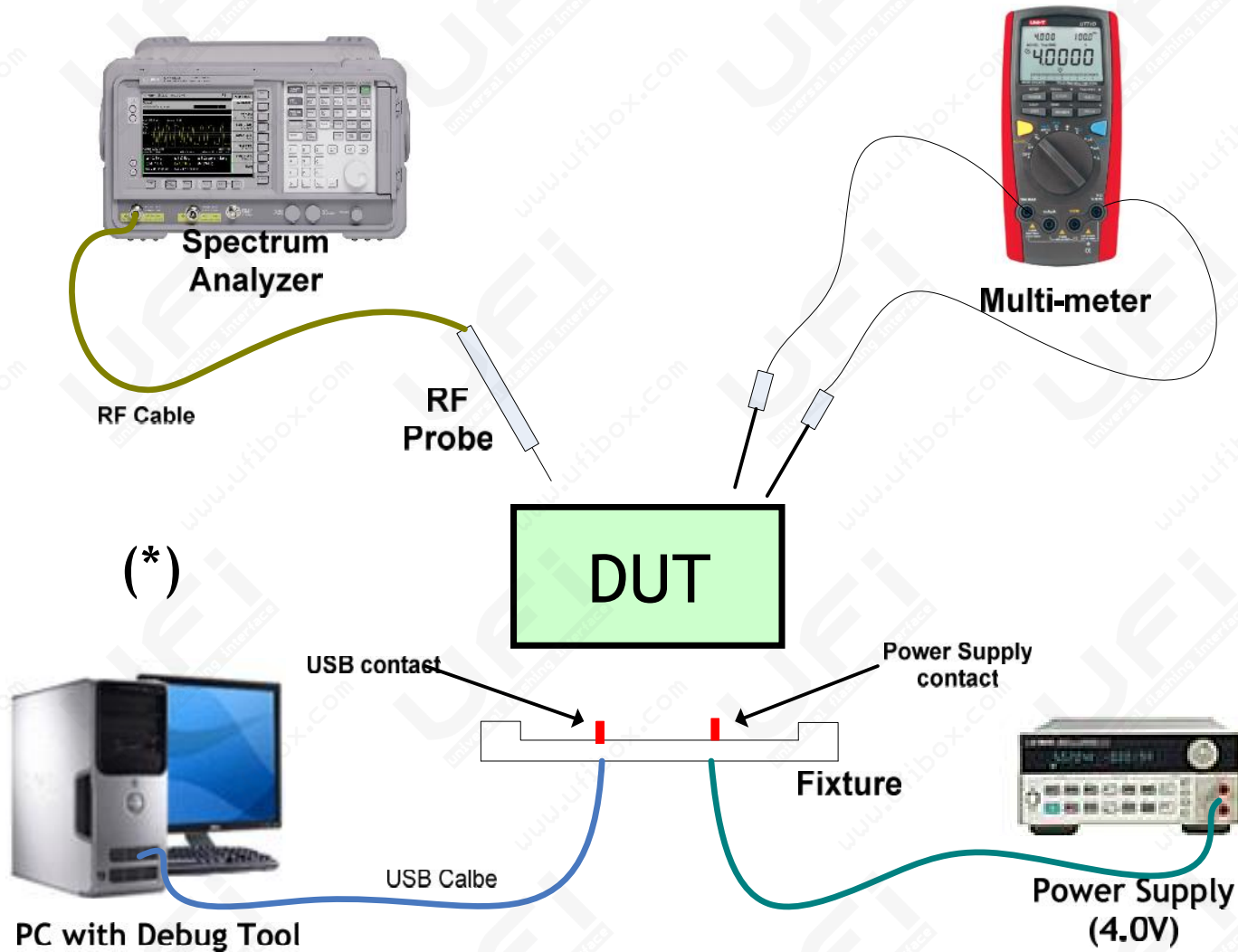
- Repair and Test Environment (*)
- RF Block Diagram
- RF function Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band8
 - LTE Quad-bands Block Diagram
 - LTE Band3/Band7/Band8/Band20/Band28
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band3/Band7/Band8/Band20/Band28
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFi Repair Guide

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Repair and Test Environment

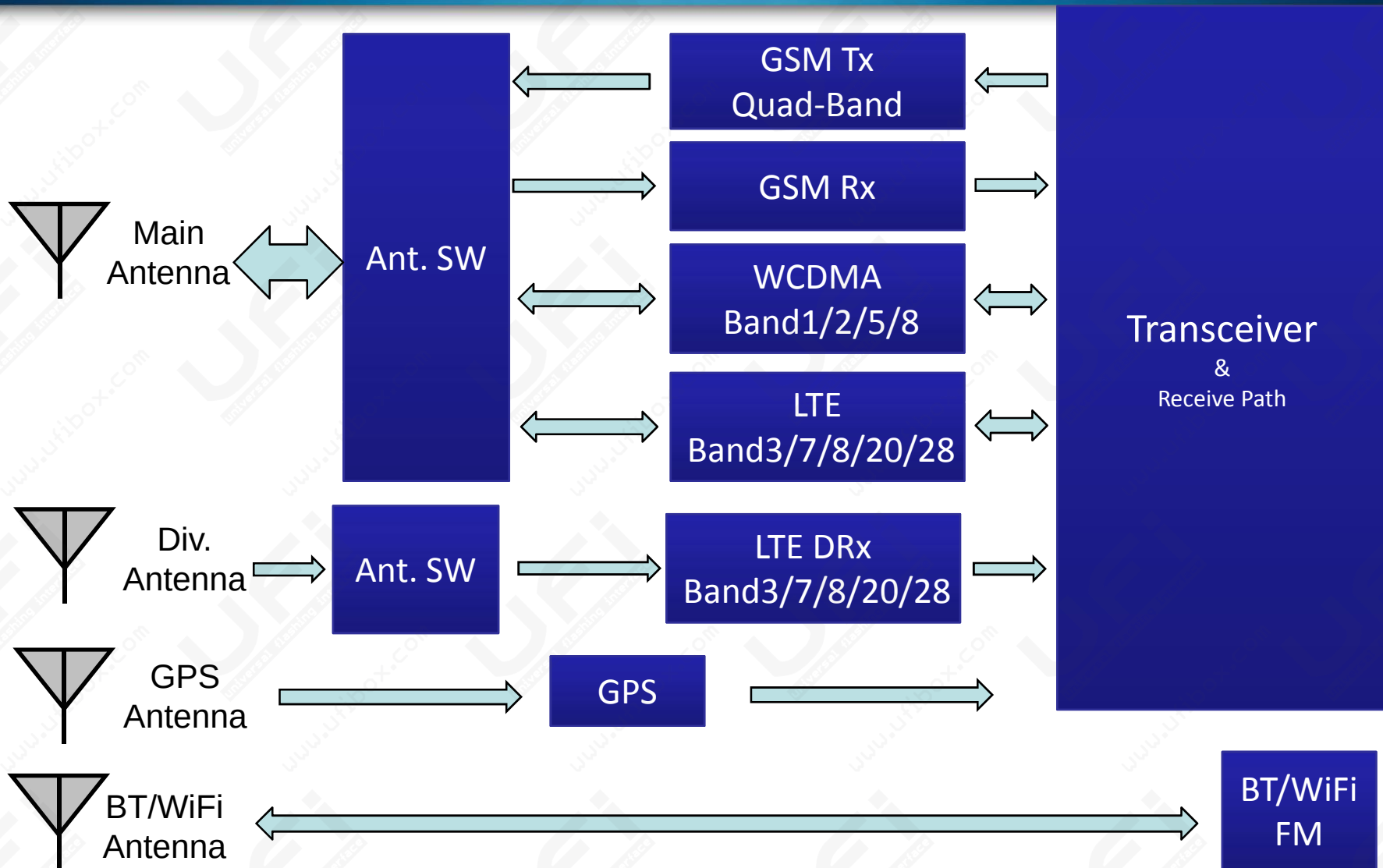


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- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
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RF Block Diagram



Sku table

TW&JP sku

GSM	Qual Band
WCDMA	Qual Band
FDD-LTE	B1/B2/B3/B5/B6/B8/B9/B18/B19/B28

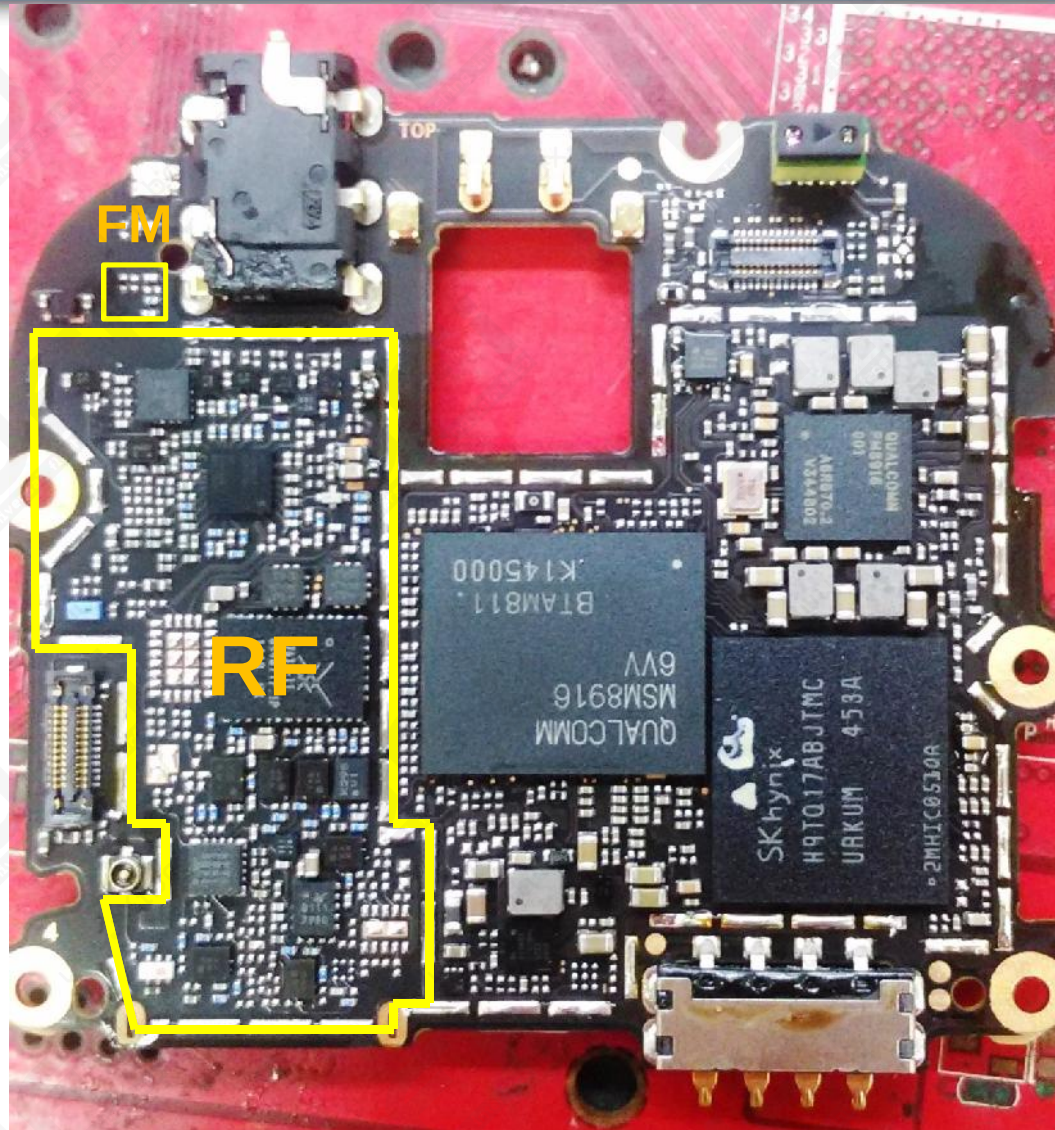
WW sku

GSM	Qual Band
WCDMA	Qual Band
FDD-LTE	B1/B2/B3/B5/B7/B8/B20

CUCC sku

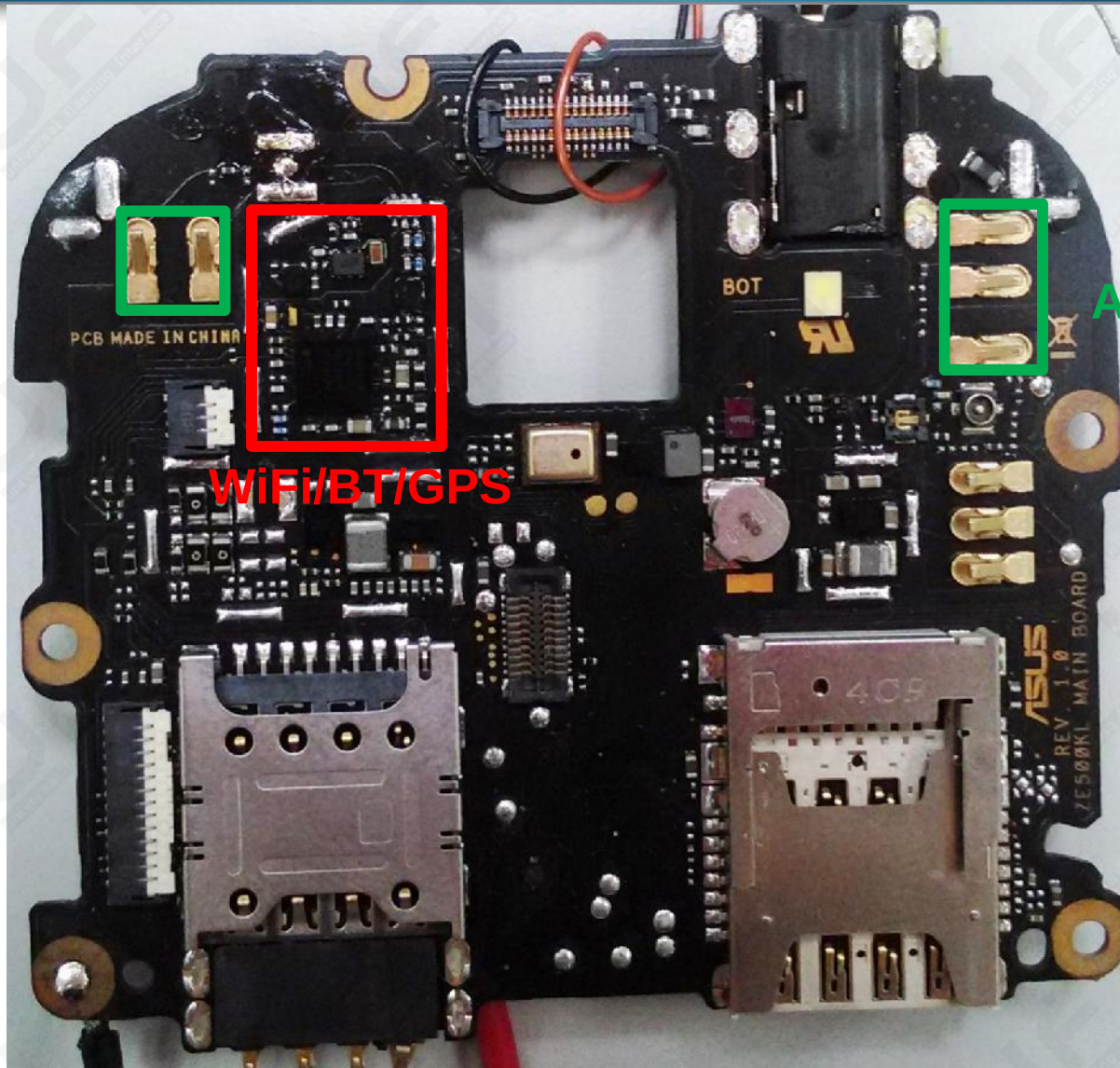
GSM	Qual Band
WCDMA	Qual Band
FDD-LTE	B1/B3
TDD-LTE	B38/B40/B41

ZE500KL RF Areas Top Side



ZE500KL RF Areas Bottom Side

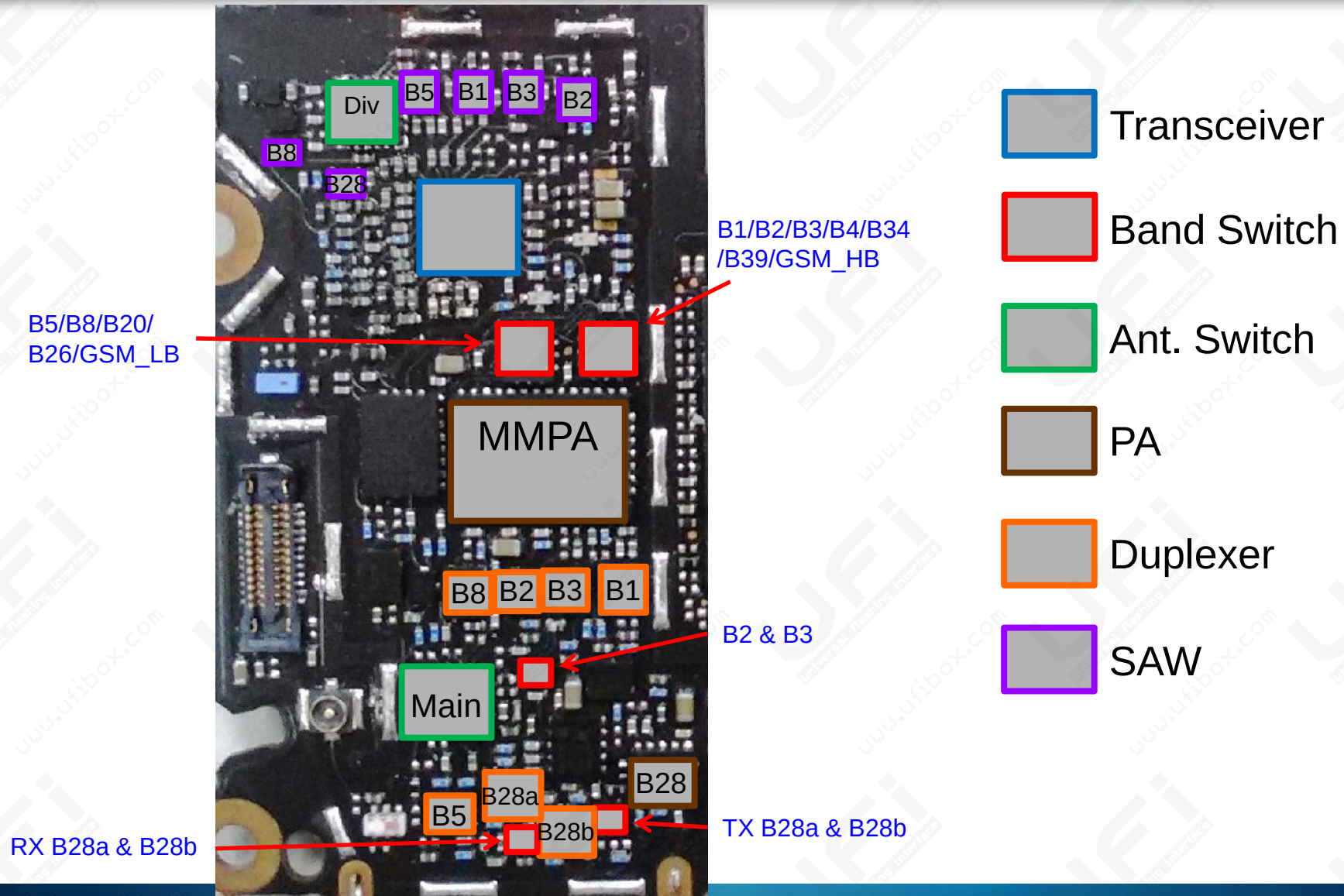
WiFi/BT GPS
Ant. Spring



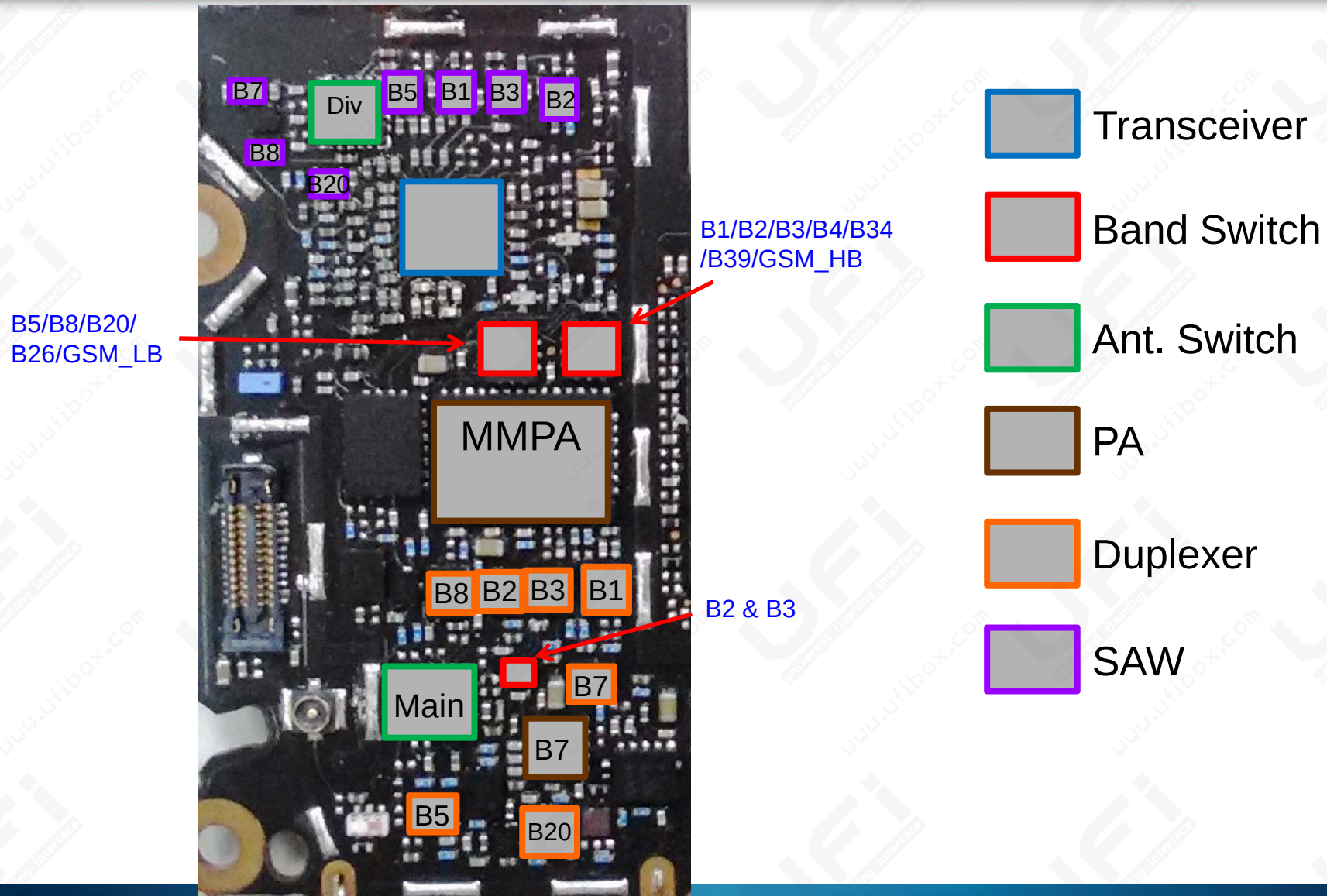
4G Div.
Ant. Spring

WiFi/BT/GPS

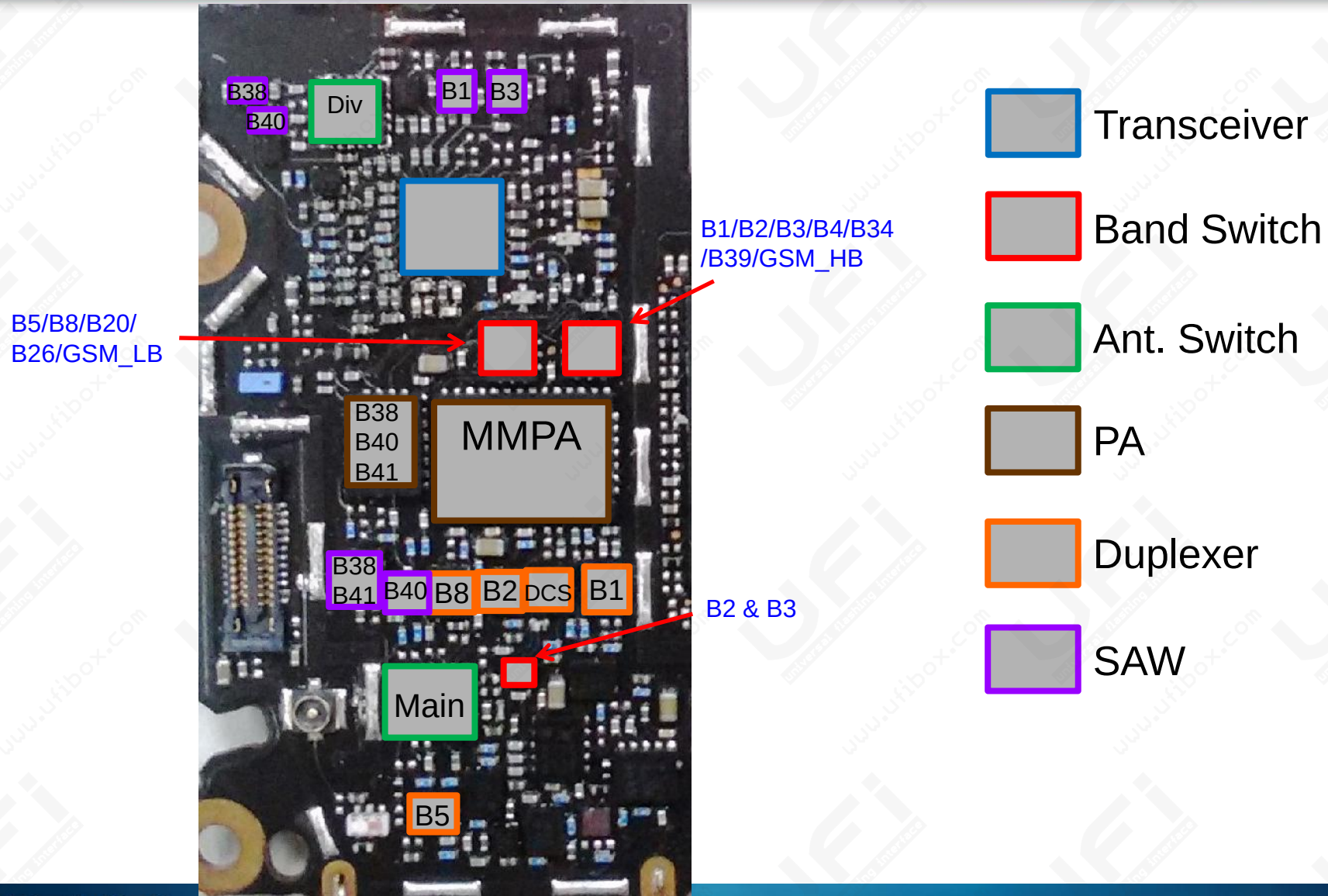
TW sku key components locations(Top)



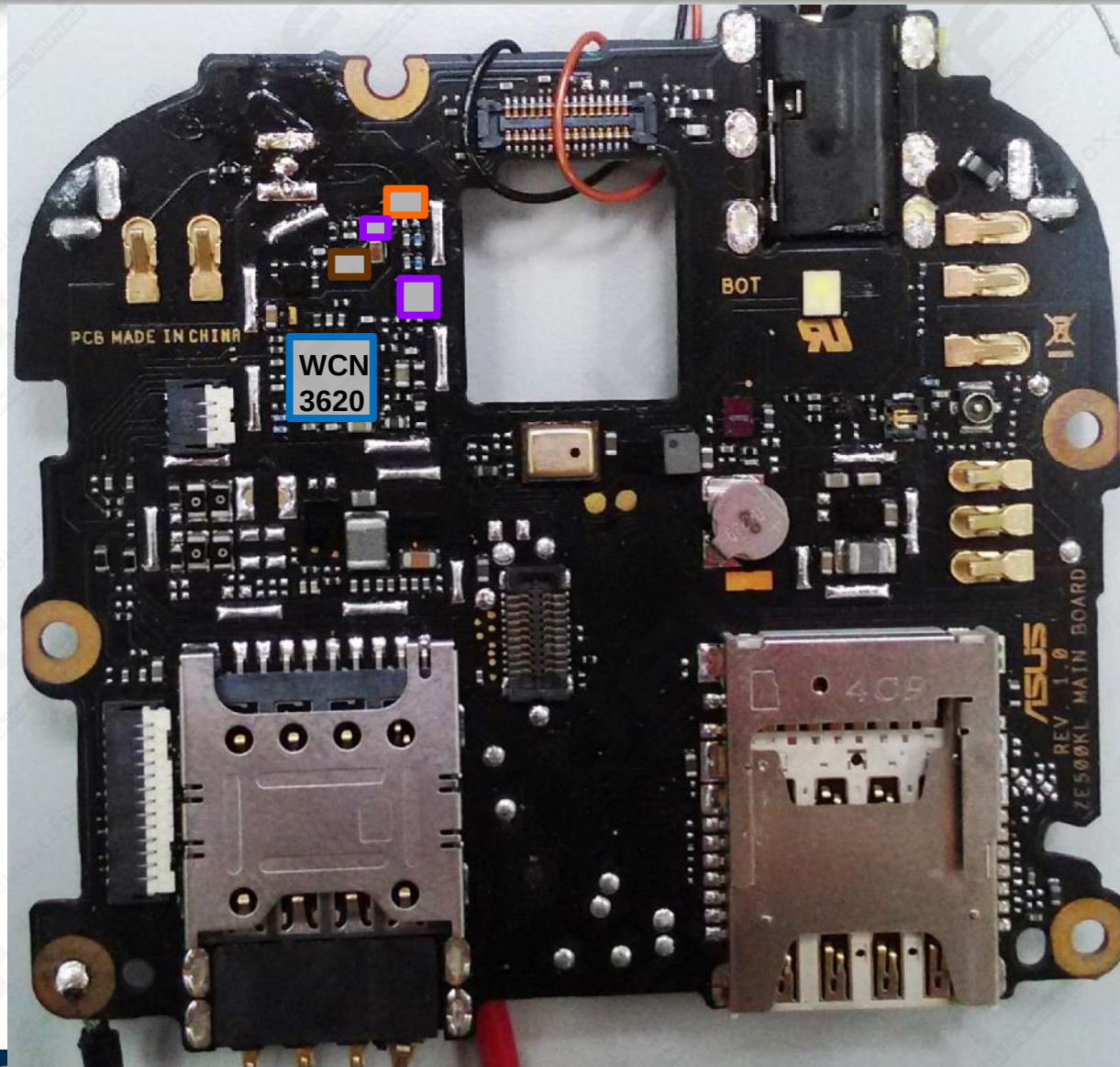
WW sku key components locations(Top)

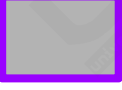


CUCC key components locations(Top)



Key components locations(Bot)



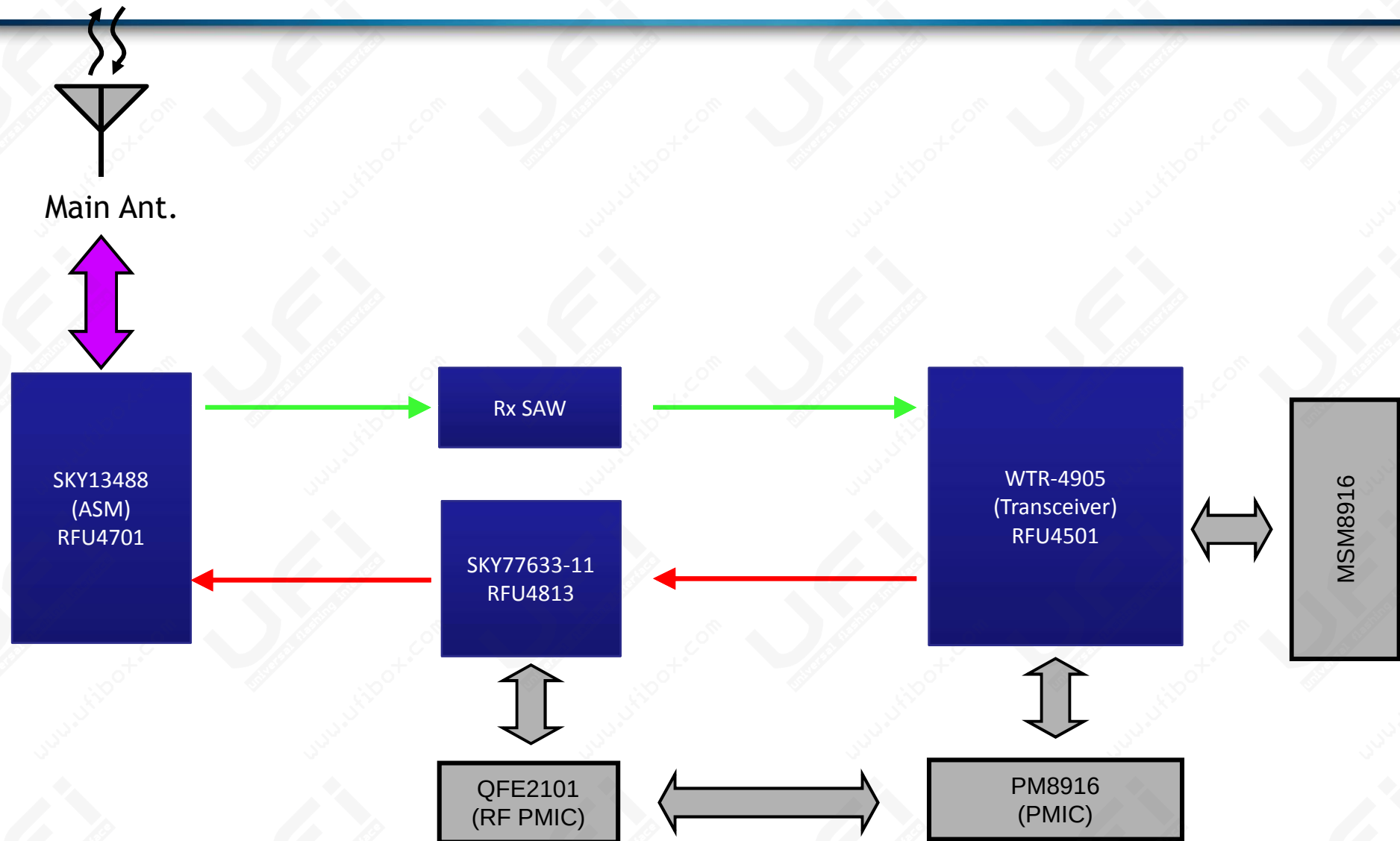
-  LNA
-  Diplexer
-  SAW

Outline

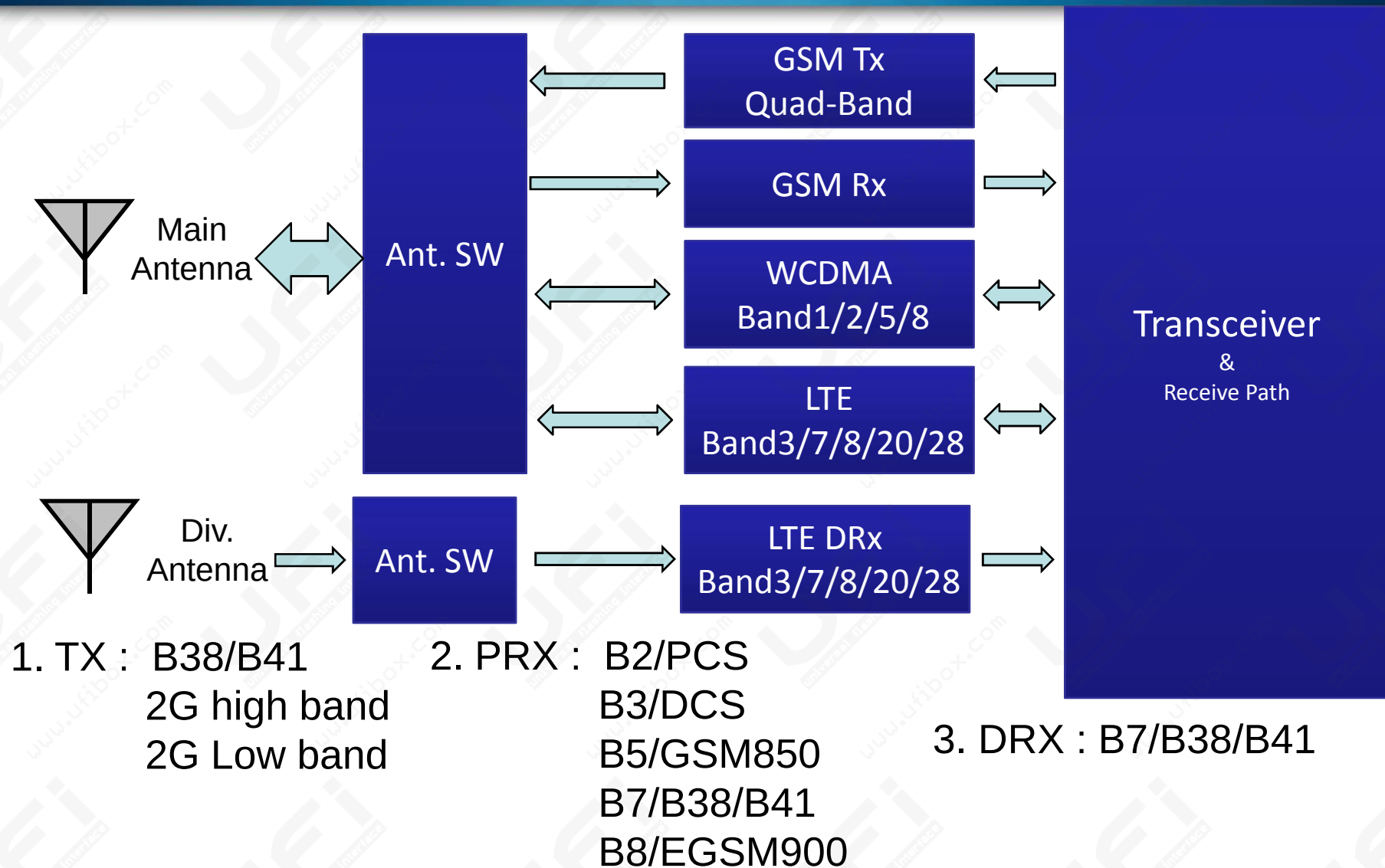
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GSM Block Diagram

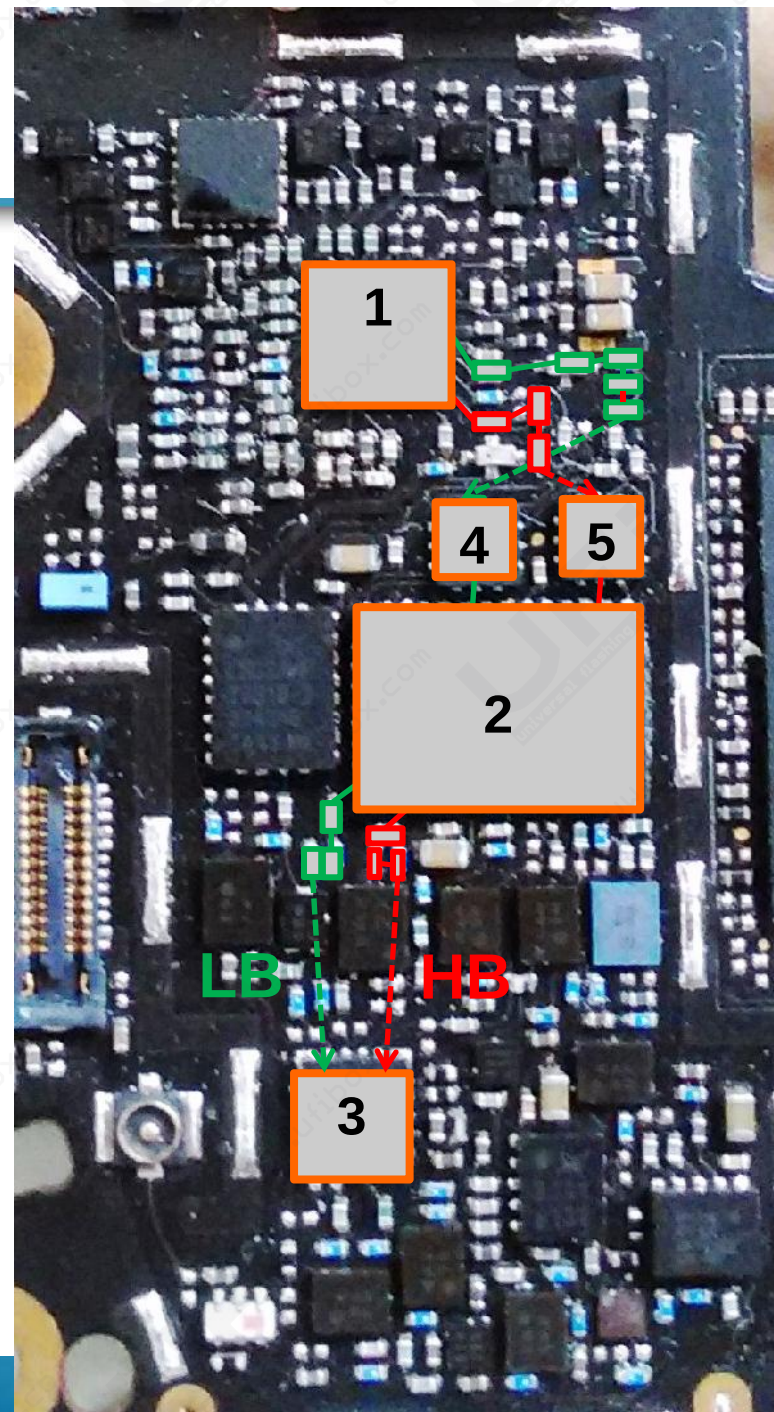
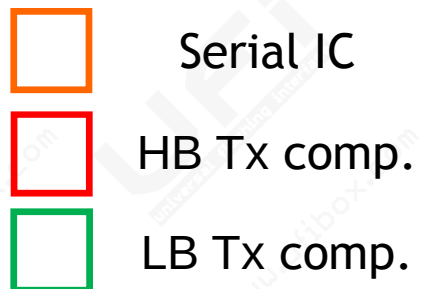


Use the same path



2G TX

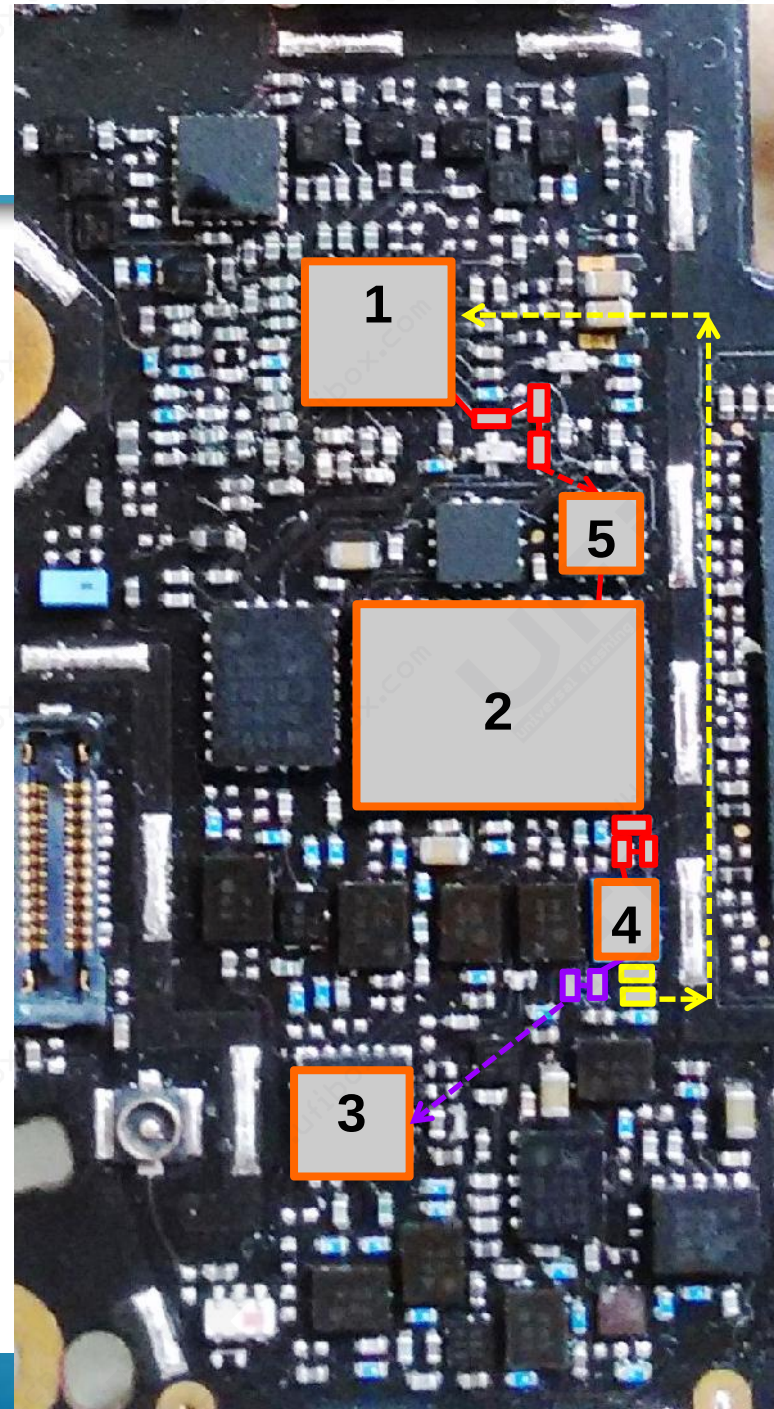
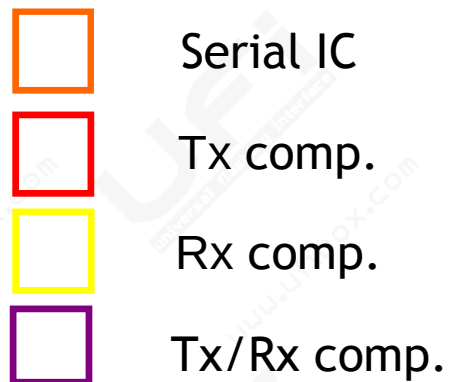
1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4811 : Band Switch
5. RFU4812 : Band Switch



Top

WCDMA/LTE B1 TX/RX

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4801 : B1 Duplexer
5. RFU4811 : Band Switch

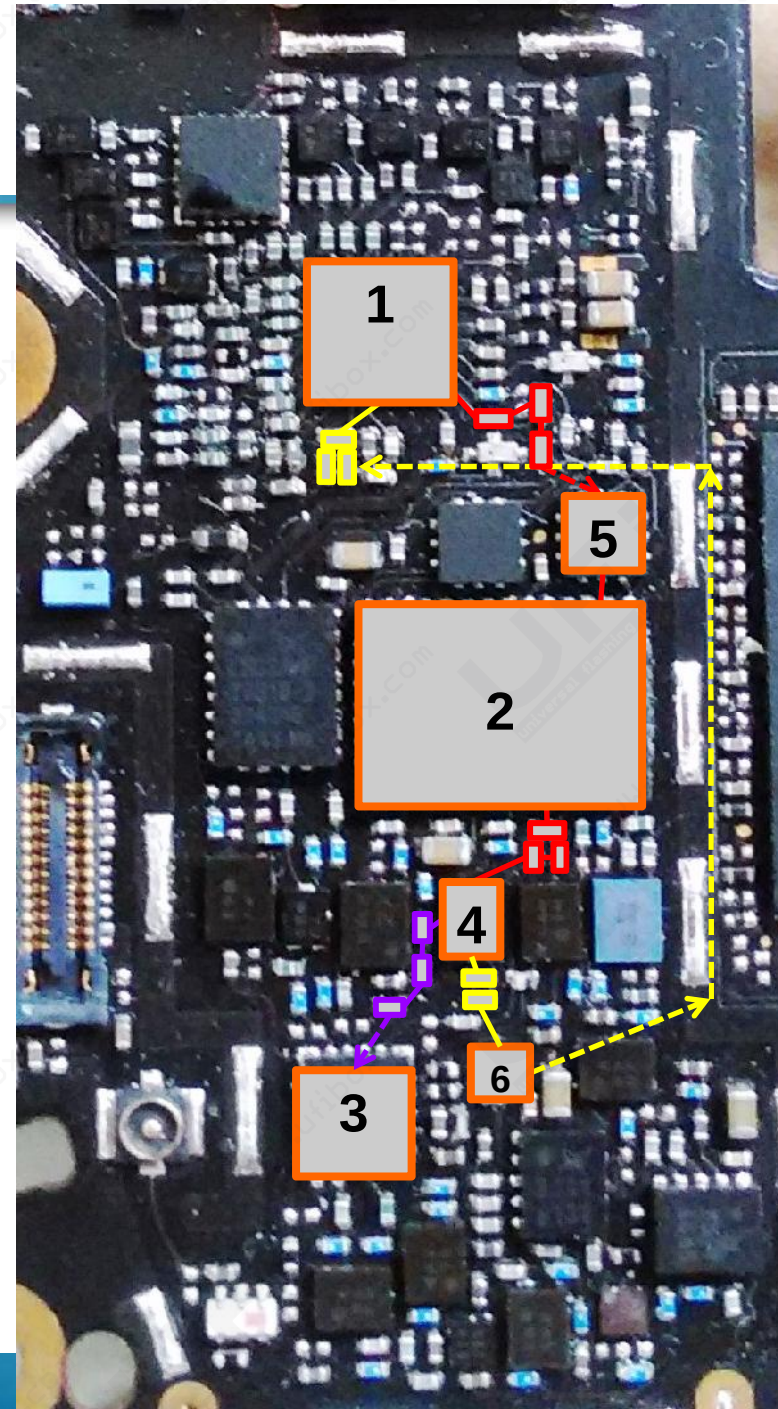
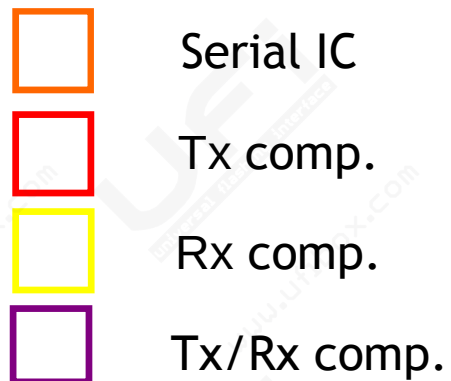


Top

B2-----TX/RX PCS -----RX

Top

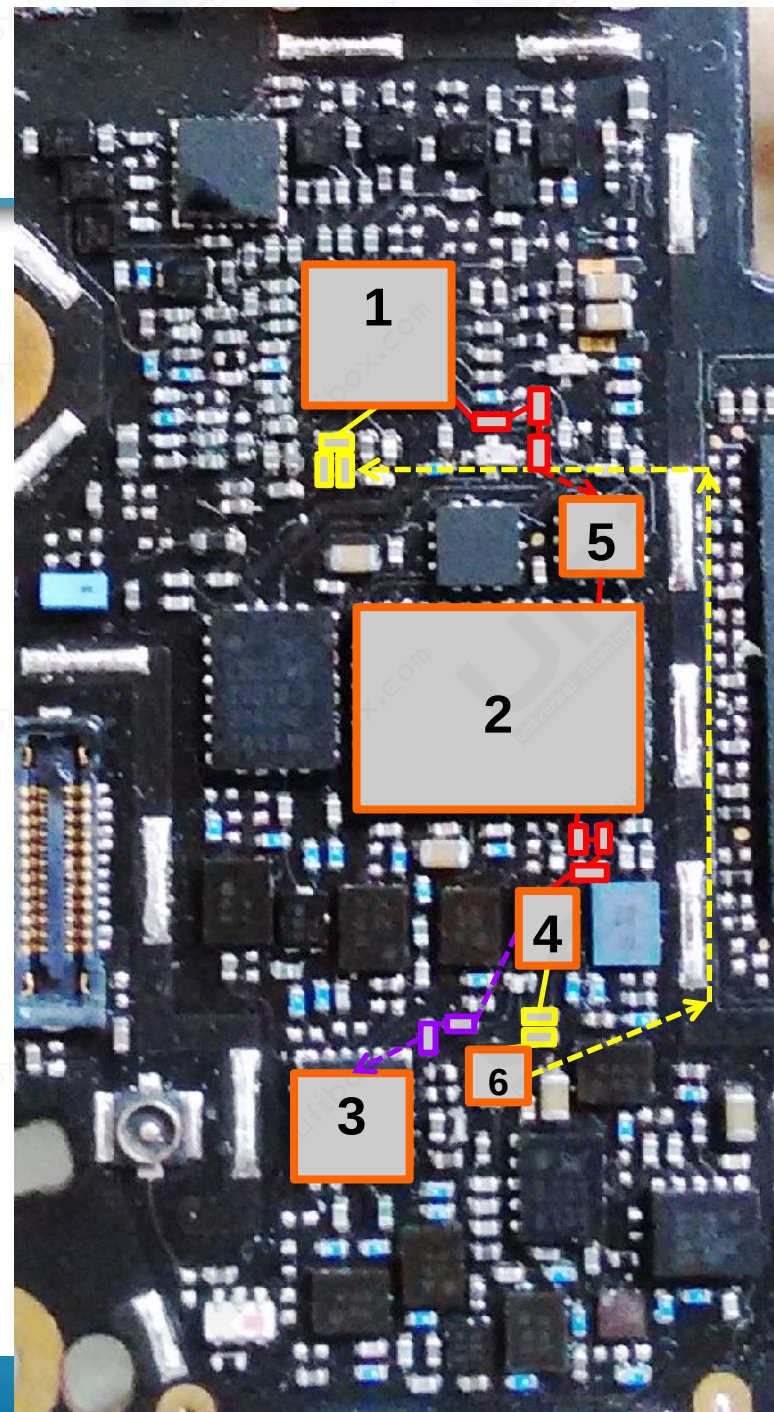
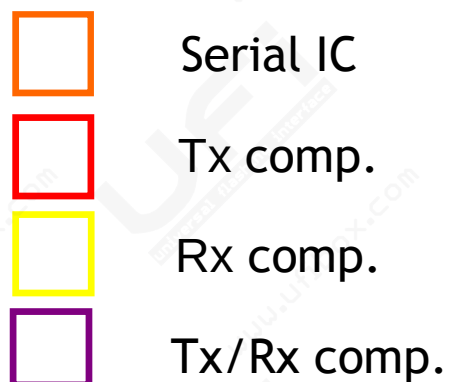
1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4805 : B2 Duplexer
5. RFU4811 : Band Switch
6. RFU4809 : Band Switch



B3/B9-----TX/RX DCS -----RX

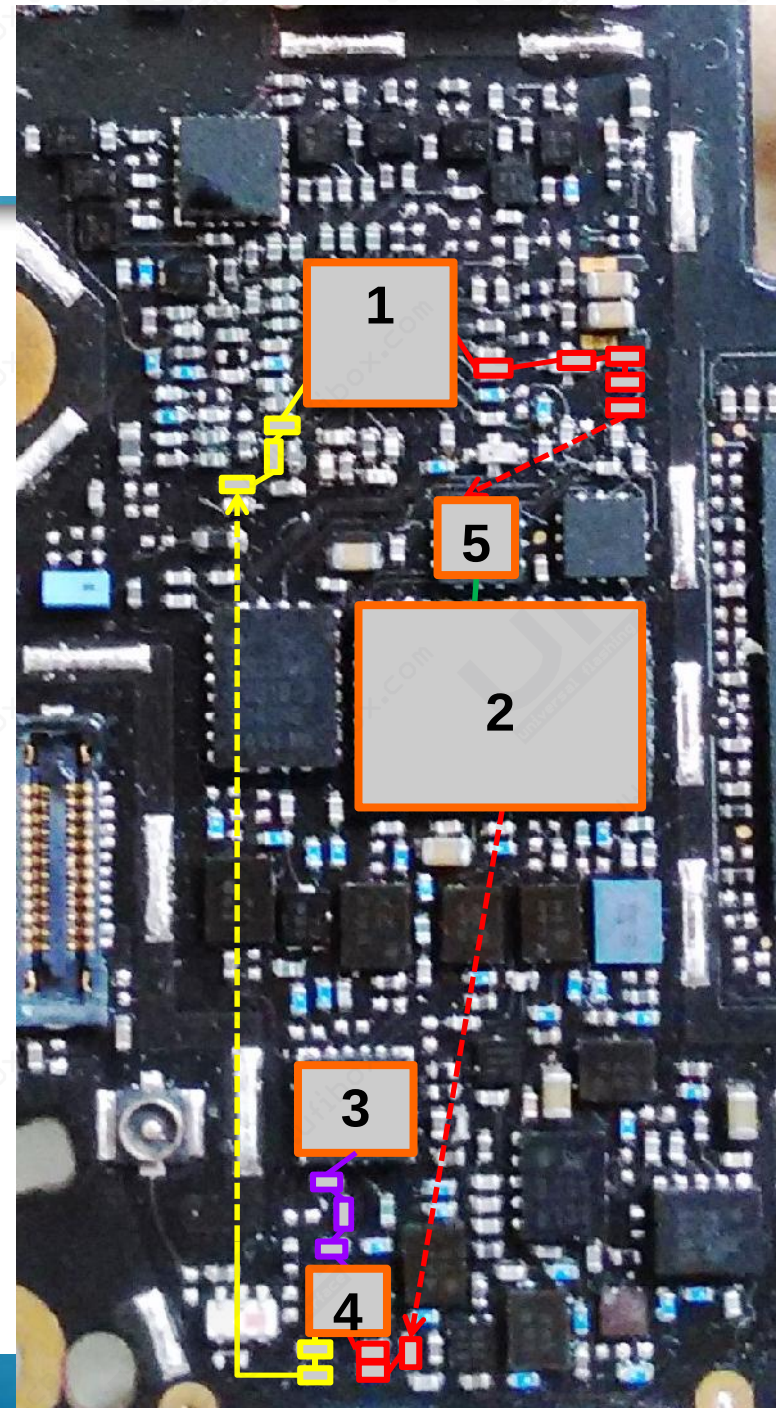
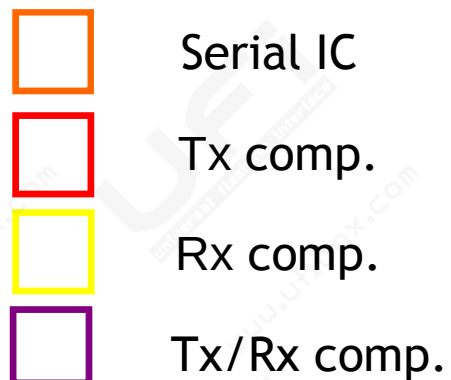
Top

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4802 : B3 Duplexer
5. RFU4811 : Band Switch
6. RFU4809 : Band Switch



B5/6/18/19----- TX/RX GSM850 ----- RX

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4803 : B5 Duplexer
5. RFU4812 : Band Switch

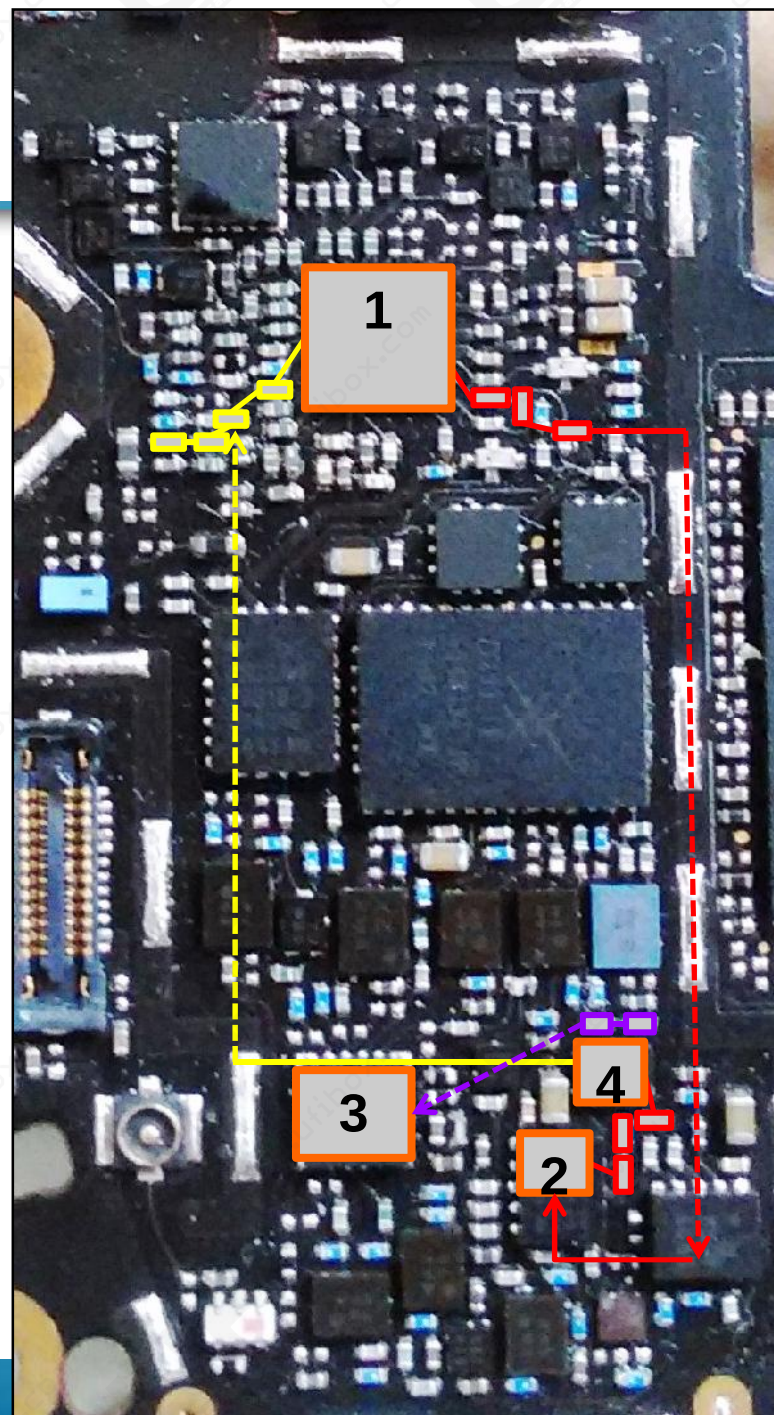
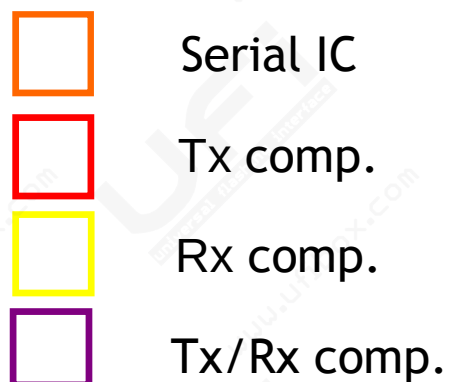


Top

B7 ----- TX/RX

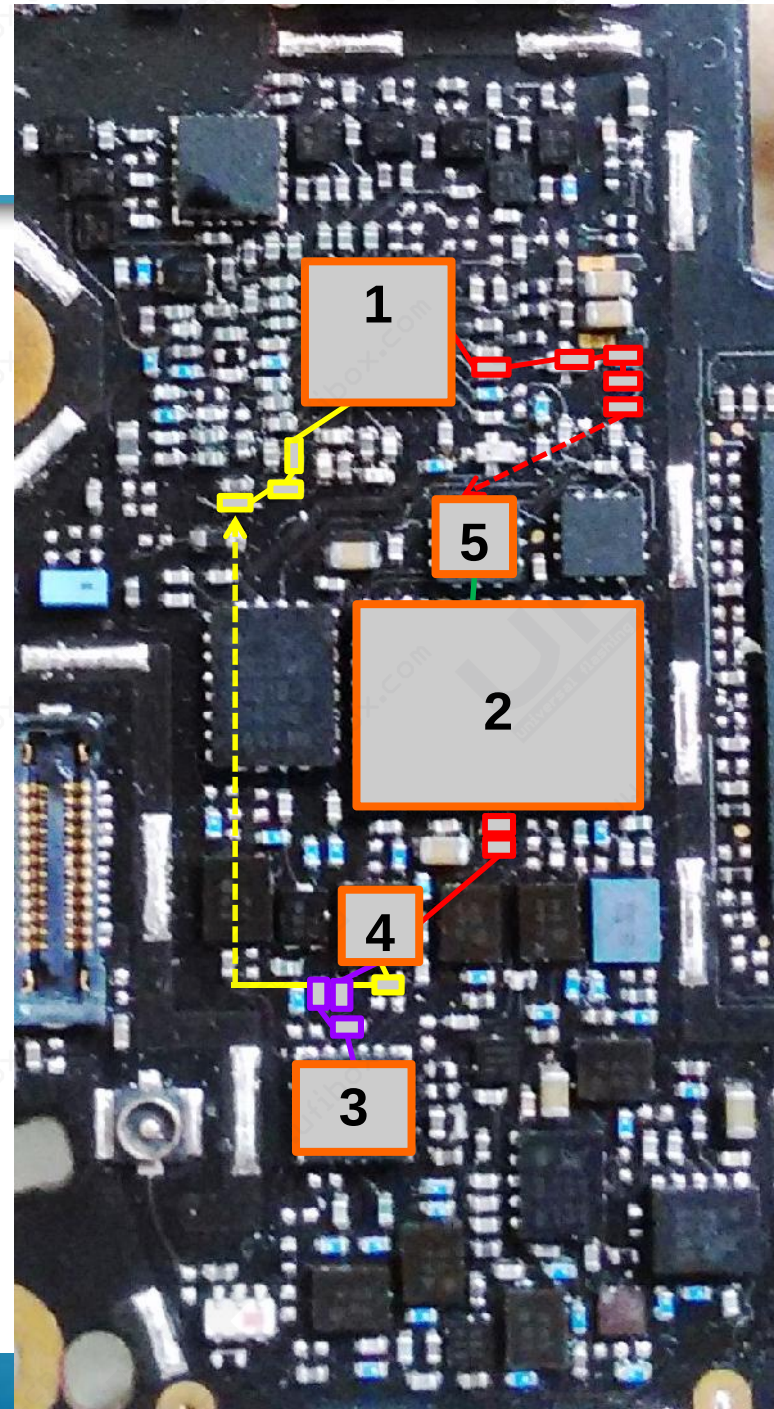
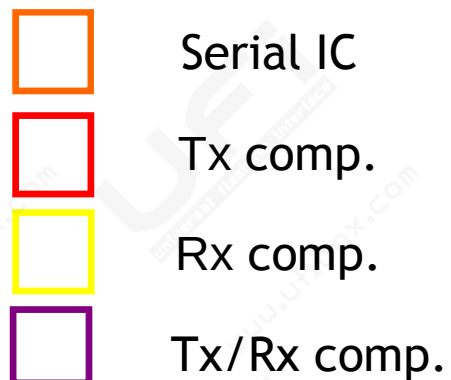
Top

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
\$SKY77781-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4803 : B7 Duplexer



B8 ----- TX/RX EGSM900 ----- RX

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU4803 : B8 Duplexer
5. RFU4812 : Band Switch



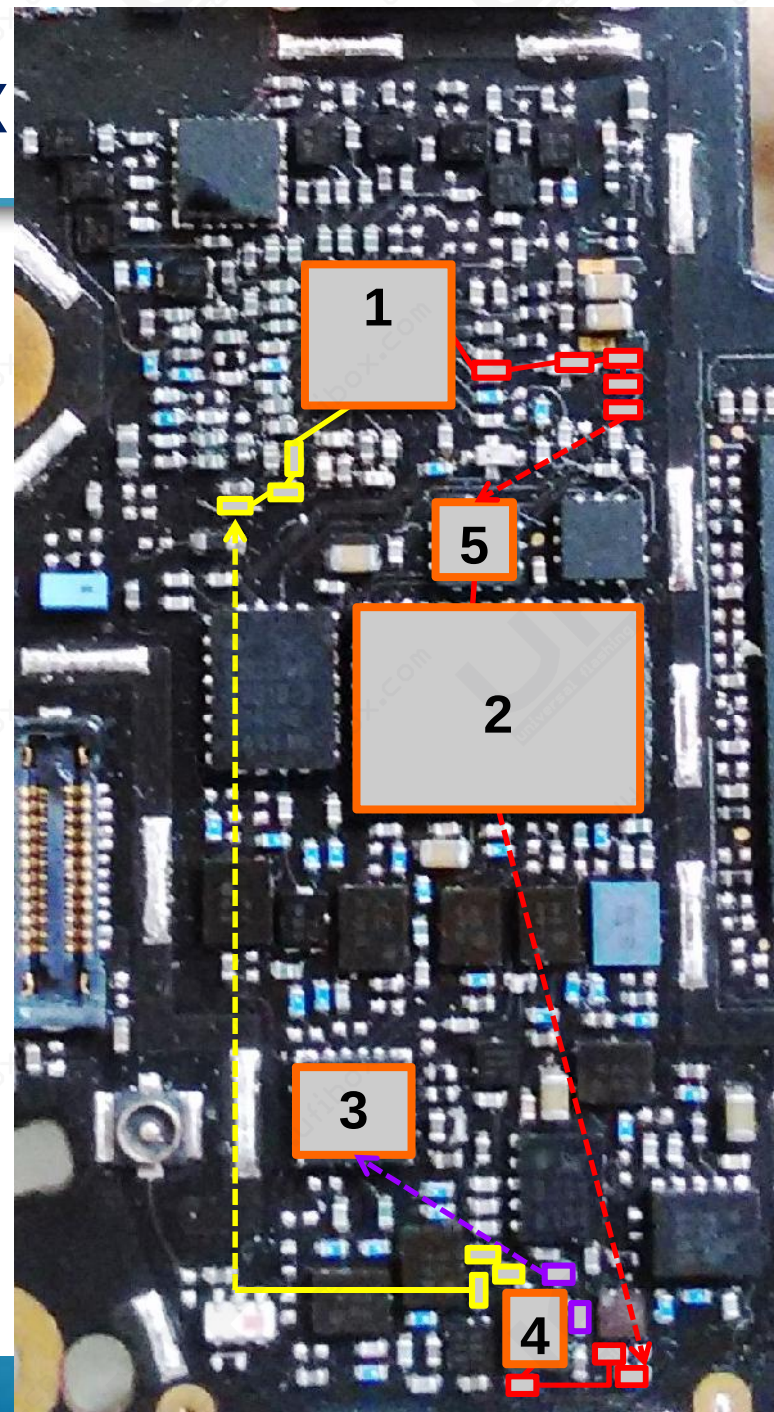
Top

B20 ----- TX/RX

Top

1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU5010: B20 Duplexer
5. RFU4812 : Band Switch

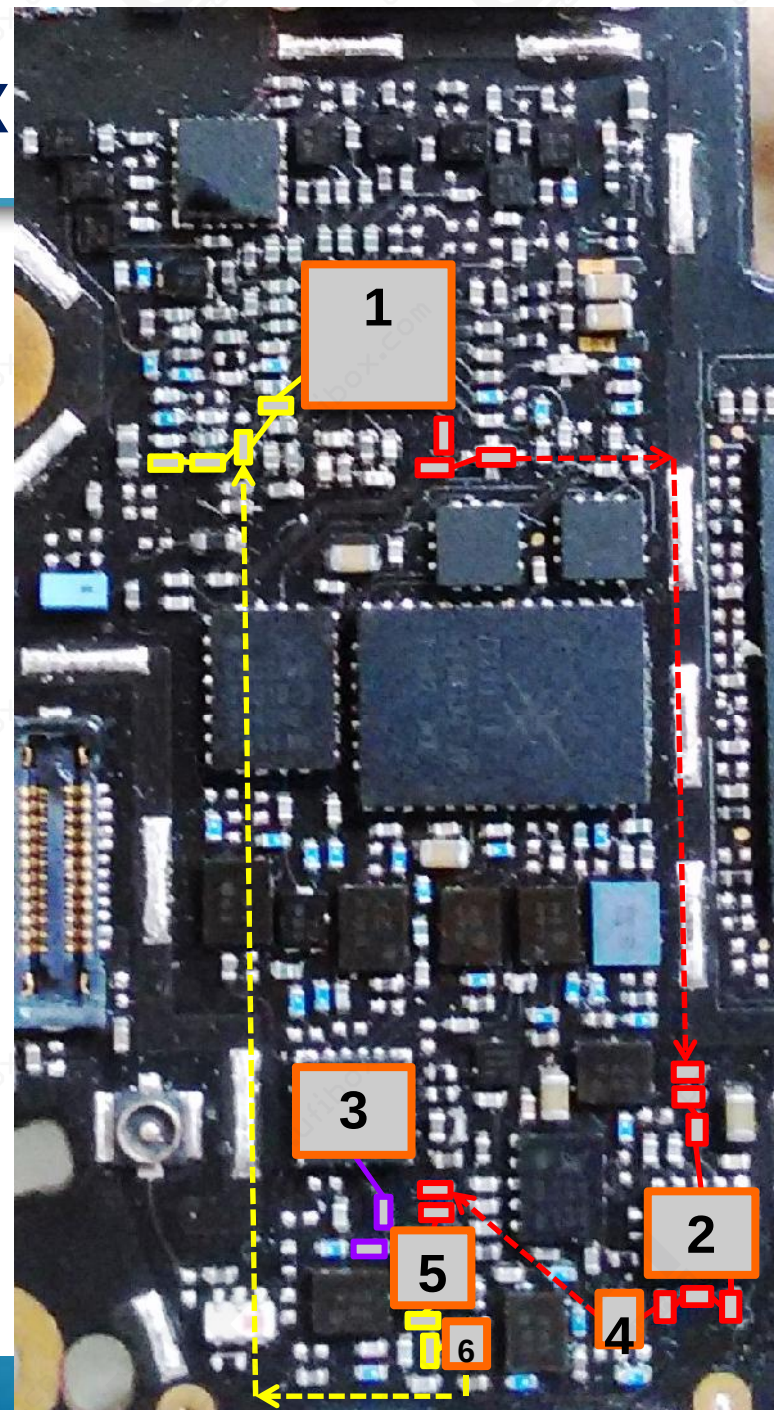
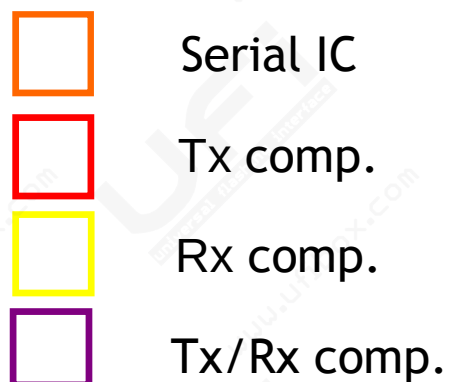
-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



B28A ----- TX/RX

Top

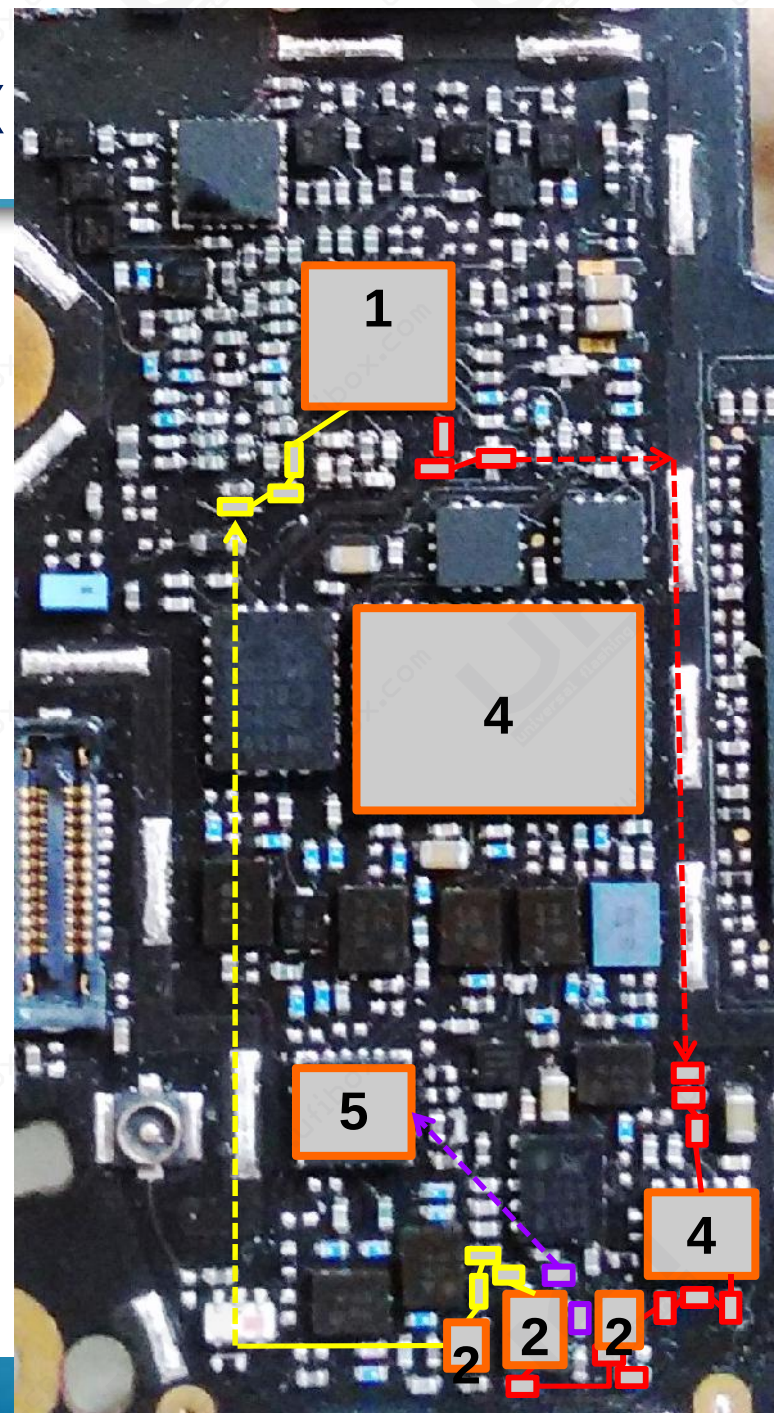
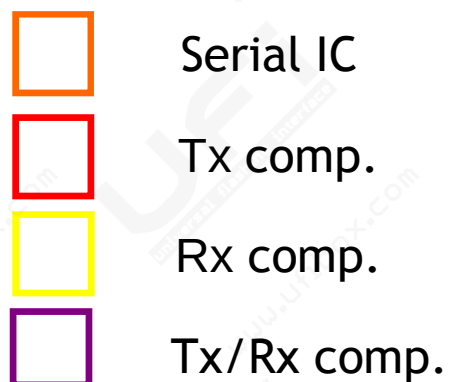
1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU5008: Band Switch
5. RFU5009: B28A Duplexer
6. RFU5011: Band Switch



B28B ----- TX/RX

Top

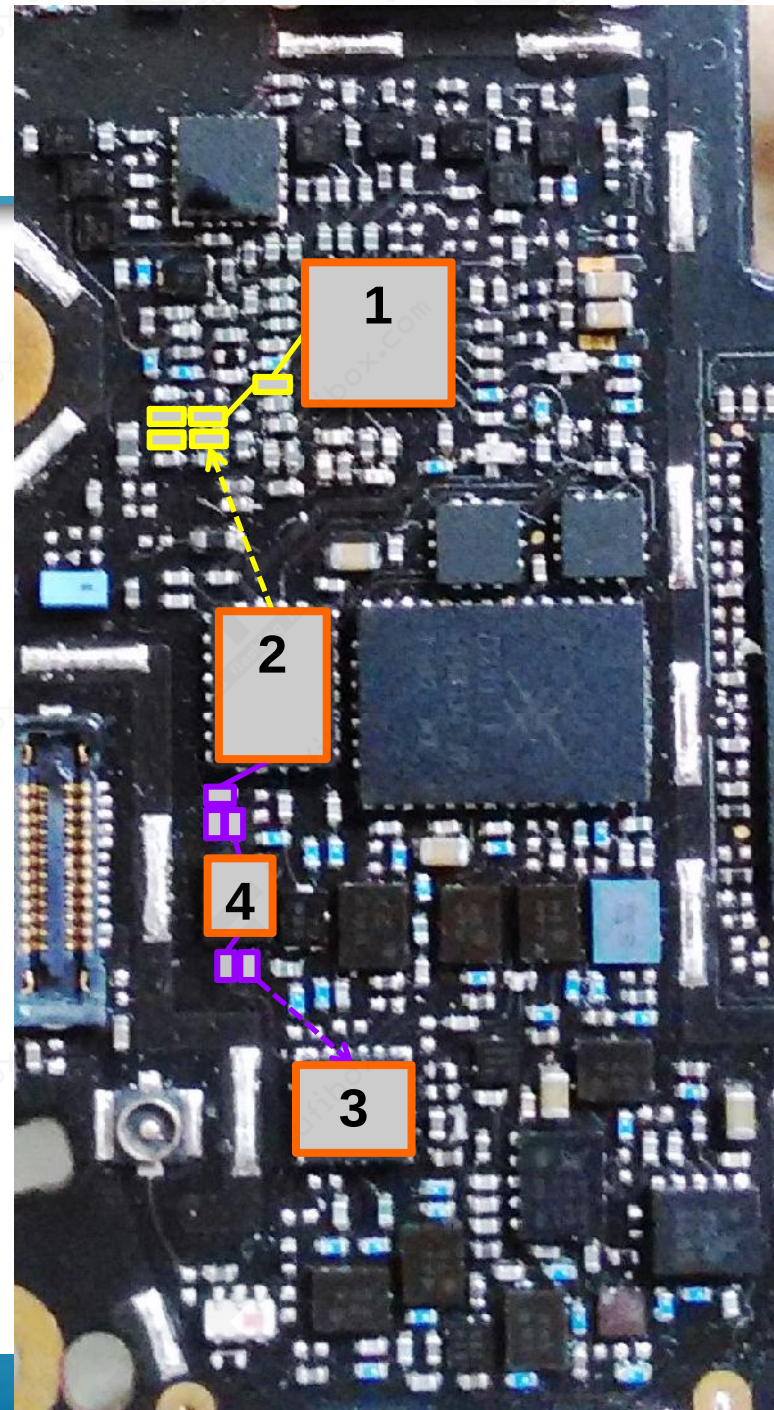
1. RFU4501: Transceiver
WTR-4905
2. RFU4813: Power Amp.
SKY77633-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU5008: Band Switch
5. RFU5010: B28B Duplexer
6. RFU5011: Band Switch



B38/B41----- TX/RX

1. RFU4501: Transceiver
WTR-4905
2. RFU5409: Power Amp.
\$SKY77818-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU5004: B38/41 Duplexer

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.

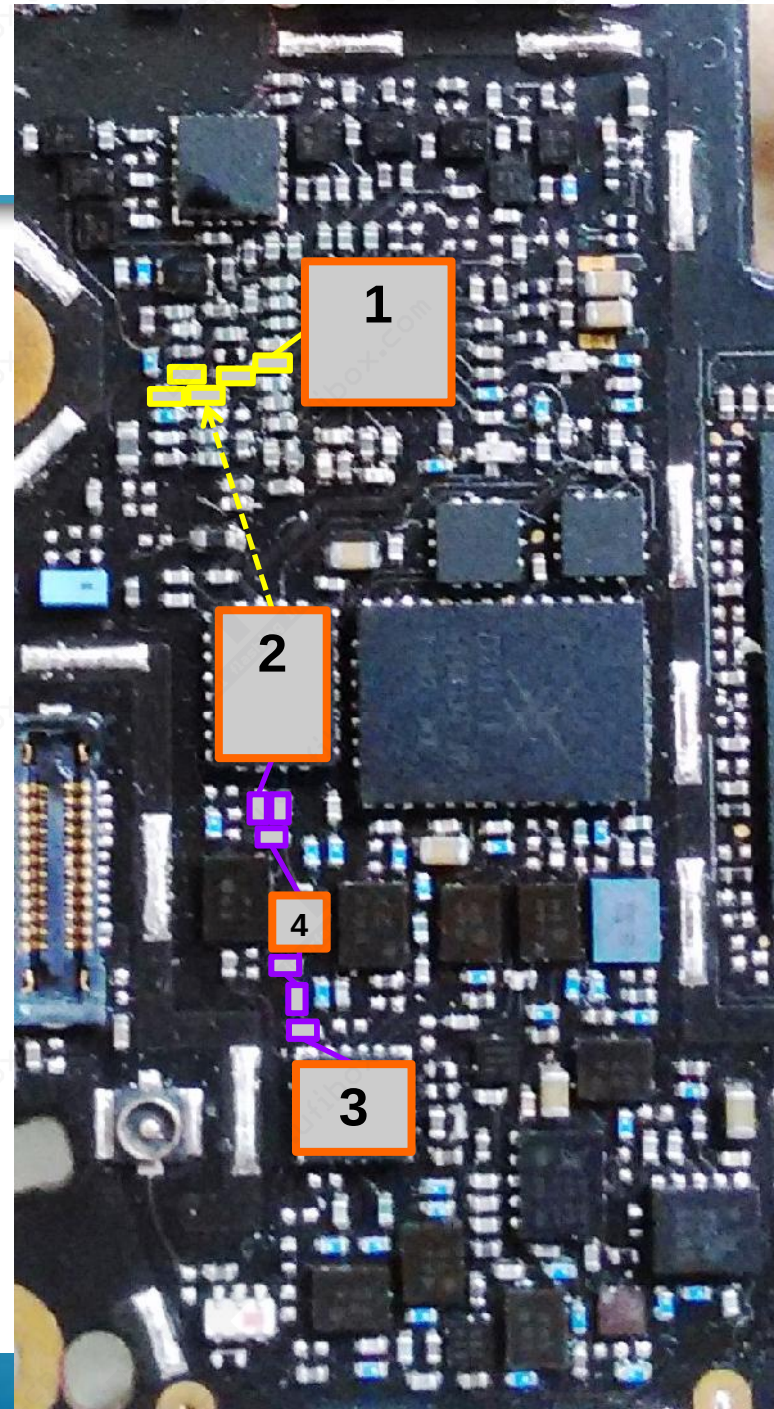


Top

B40----- TX/RX

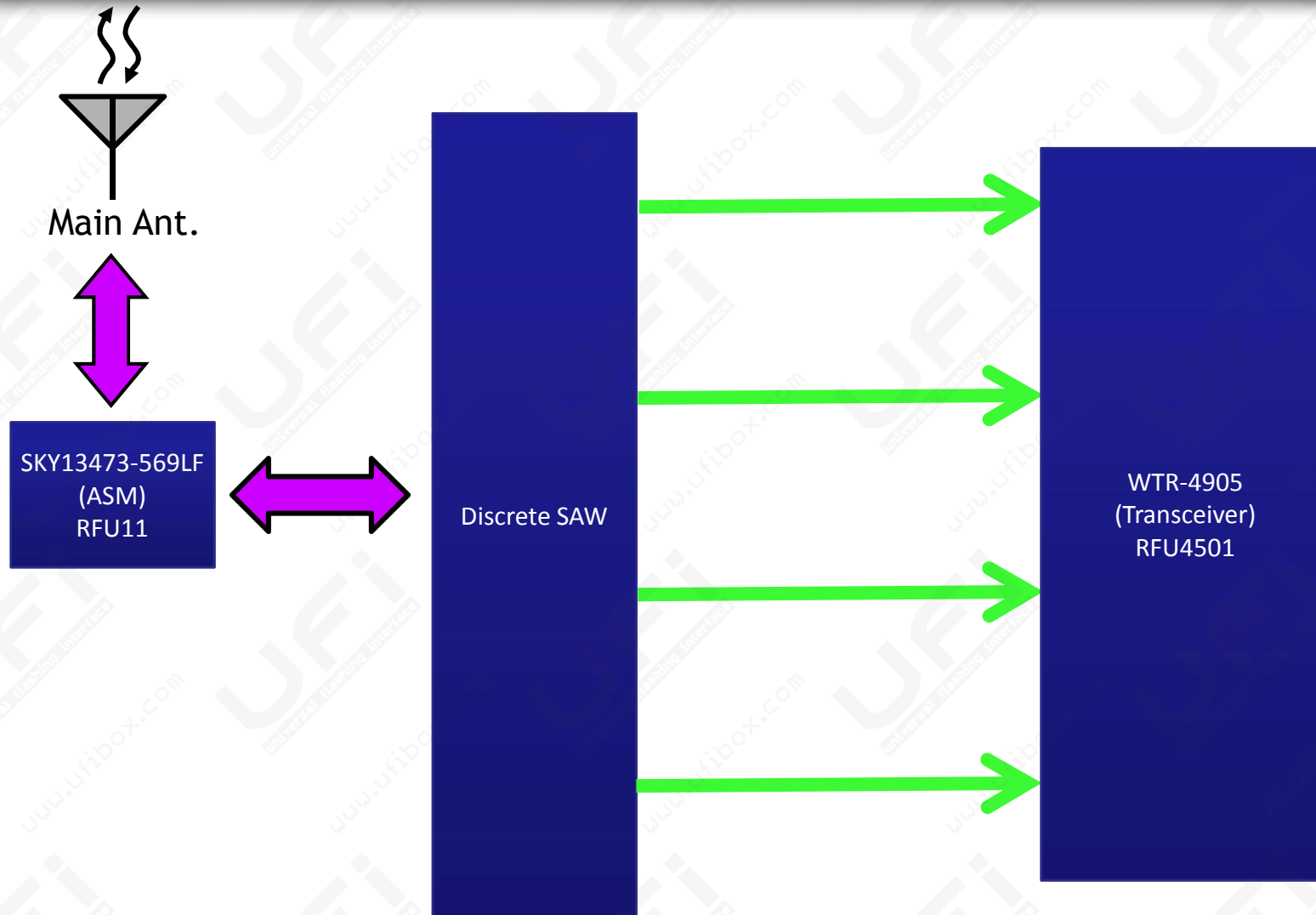
1. RFU4501: Transceiver
WTR-4905
2. RFU5409: Power Amp.
\$SKY77818-11
3. RFU4701: Main Ant. Switch
SKY13488
4. RFU5003: B38/41 Duplexer

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



Top

LTE Div. Rx Block Diagram



B1 -----DRx

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5202: B1 SAW

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



B2 -----DRx

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5208: B2 SAW
4. RFU5212: Band switch

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



B3 -----DRx

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5201: B3 SAW
4. RFU5212: Band switch

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



Band5/18/19/26 -----DRx

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5210: B5 SAW

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



Band8 -----DRx

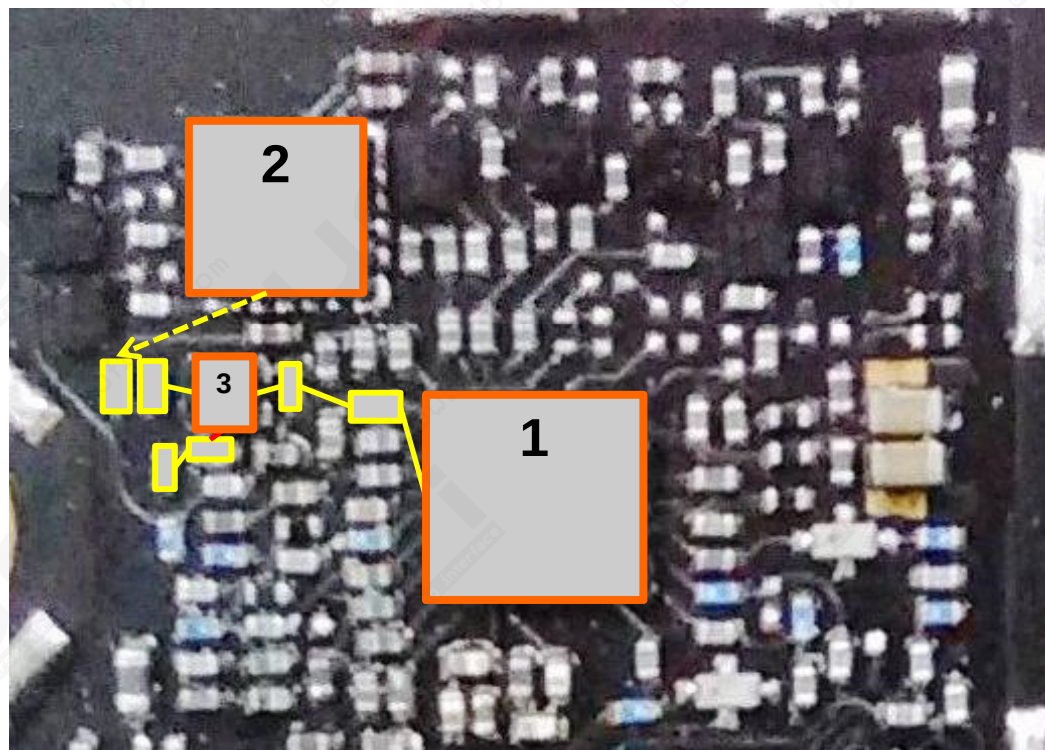
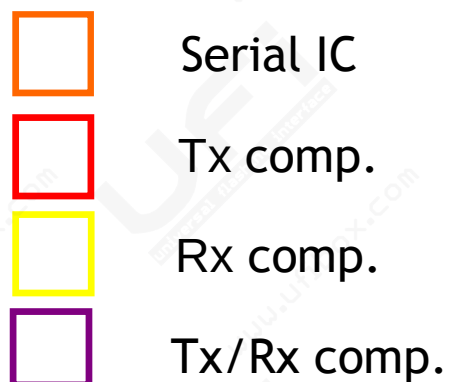
1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5209: B8 SAW

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



DRX_B28 / B20 -----DRx

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5213: B20/B28 SAW



Band40 -----DRx

Top

1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5206: B40 SAW

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.



Band38/41/7-----DRx

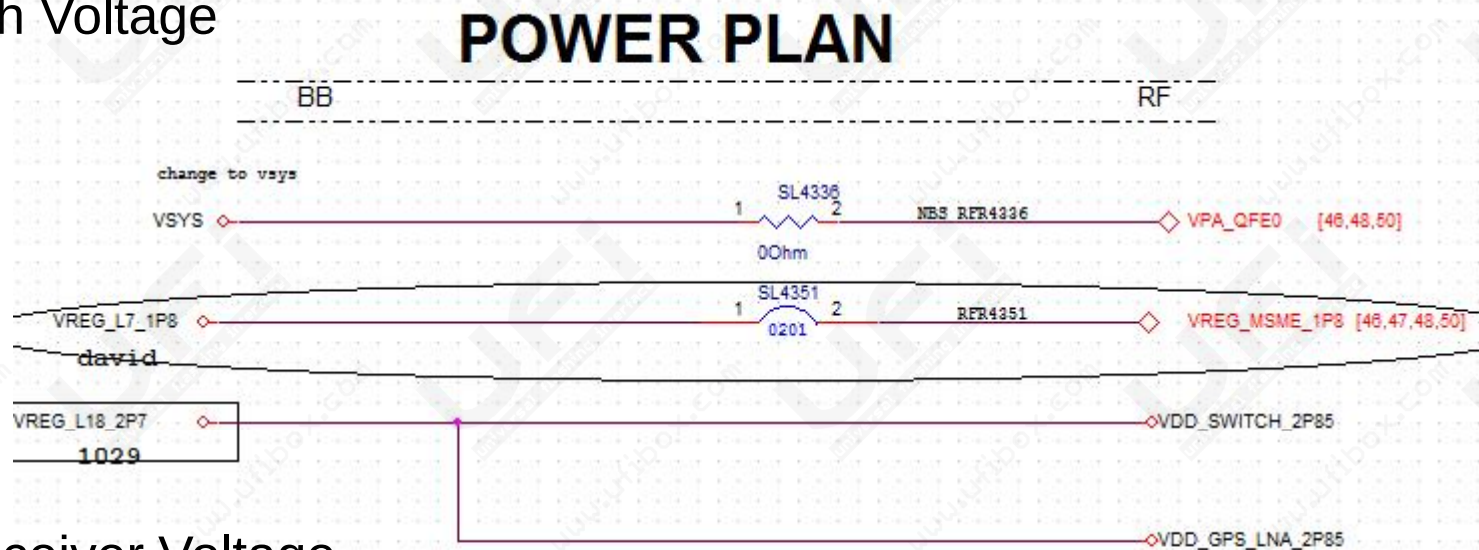
1. RFU4501: Transceiver
WTR-4905
2. RFU4702: Div. Ant. Switch
SKY13561
3. RFU5207: B38/41/7 SAW

-  Serial IC
-  Tx comp.
-  Rx comp.
-  Tx/Rx comp.

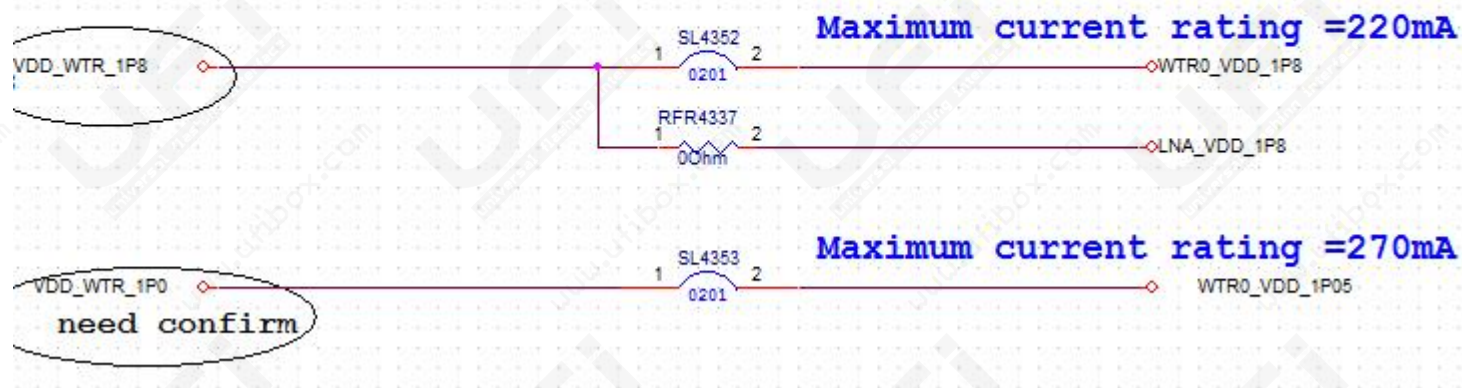


Voltage sturcutre

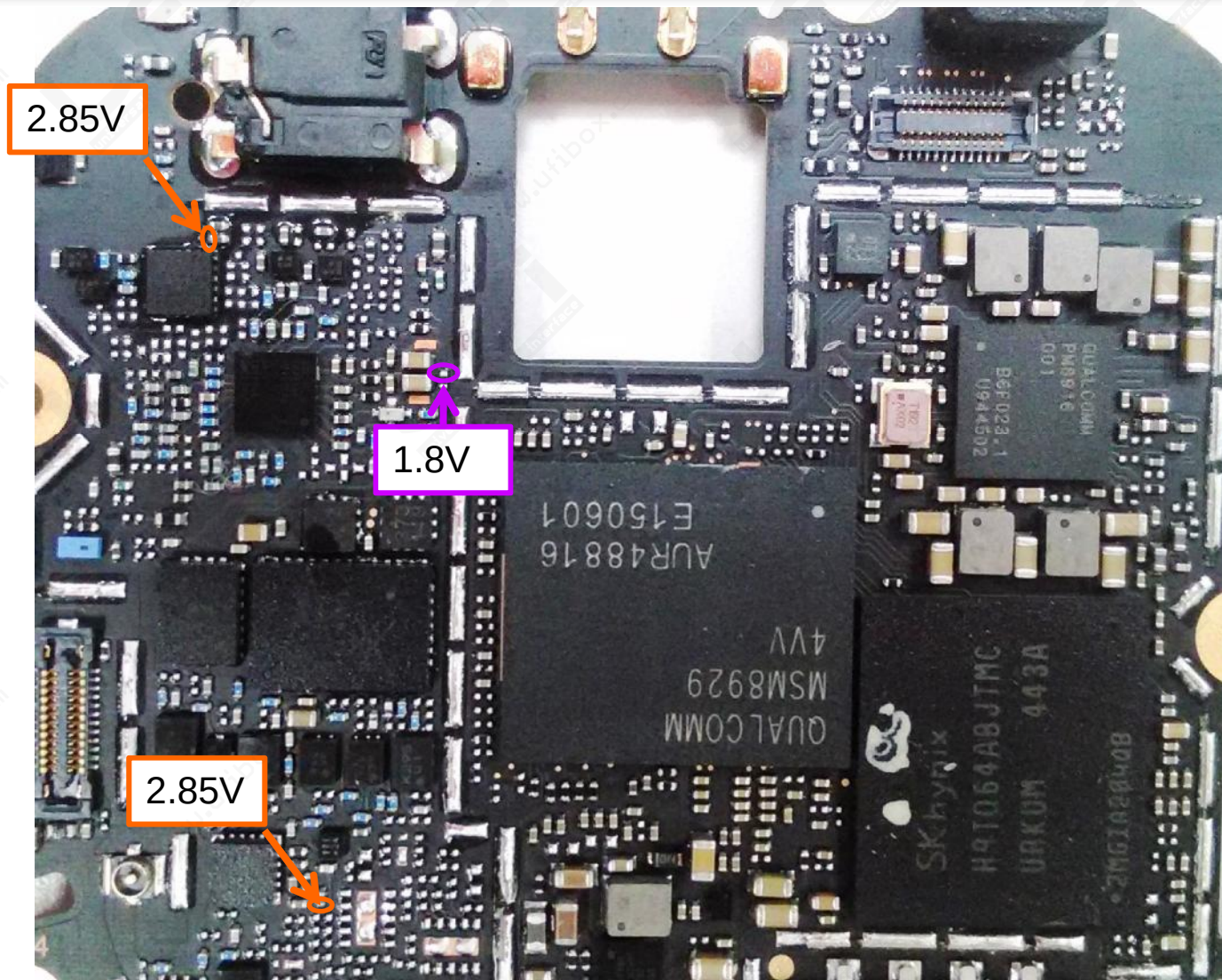
1. Switch Voltage



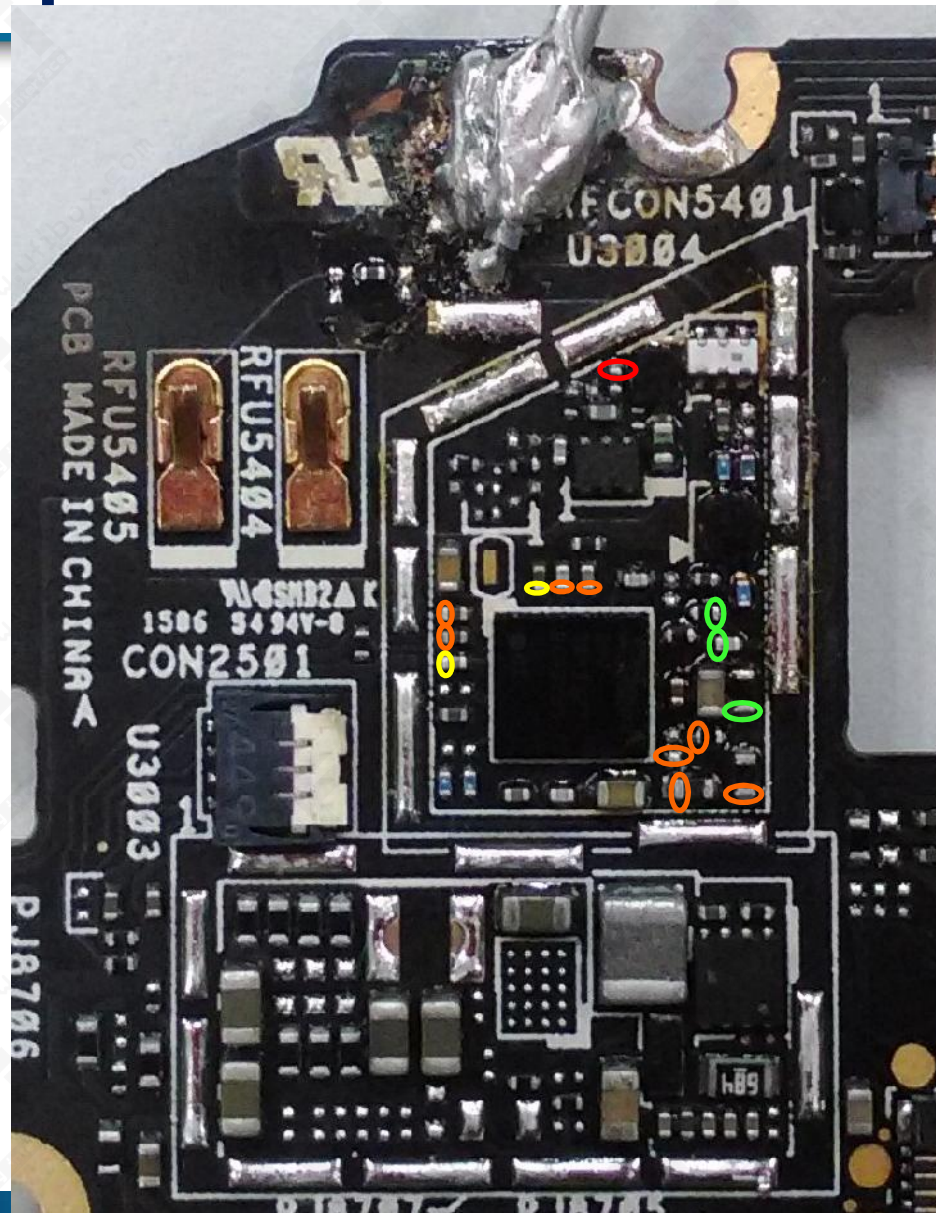
2. Transceiver Voltage



Voltage Setup



Voltage Setup



1.3V

2.85V

3.0V

1.8V

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2G RF Test

Error Code	Action
Tx Power	1. Re-Calibration 2. Check Tx Path 3. Change PA
ORFS/ Power vs. Time	1. Check Tx Path 2. Change PA
Frequency Error/ Phase Error	1. Check/change PMIC
Sensitivity/RSSI	1. Check/Change PMIC 2. Check/Change crystal 3. Check Rx Path

3G RF Test

Error Code	Action
Tx Power/ACLR/ SEM/OBW/EVM	1. Re-calibration 2. Check Tx Path 3. Change PA
Sensitivity/RSCP	1. Re-calibration 2. Check Rx Path
Frequency Error	1. Check/Change PMIC 2. Check/Change crystal

4G RF Test

Error Code	Action
Tx Power/ACLR/ SEM/OBW/EVM	1. Re-calibration 2. Check Tx Path 3. Change PA
Sensitivity/RSRP	1. Re-calibration 2. Check Rx Path
Frequency Error	1. Check/Change PMIC 2. Check/Change crystal

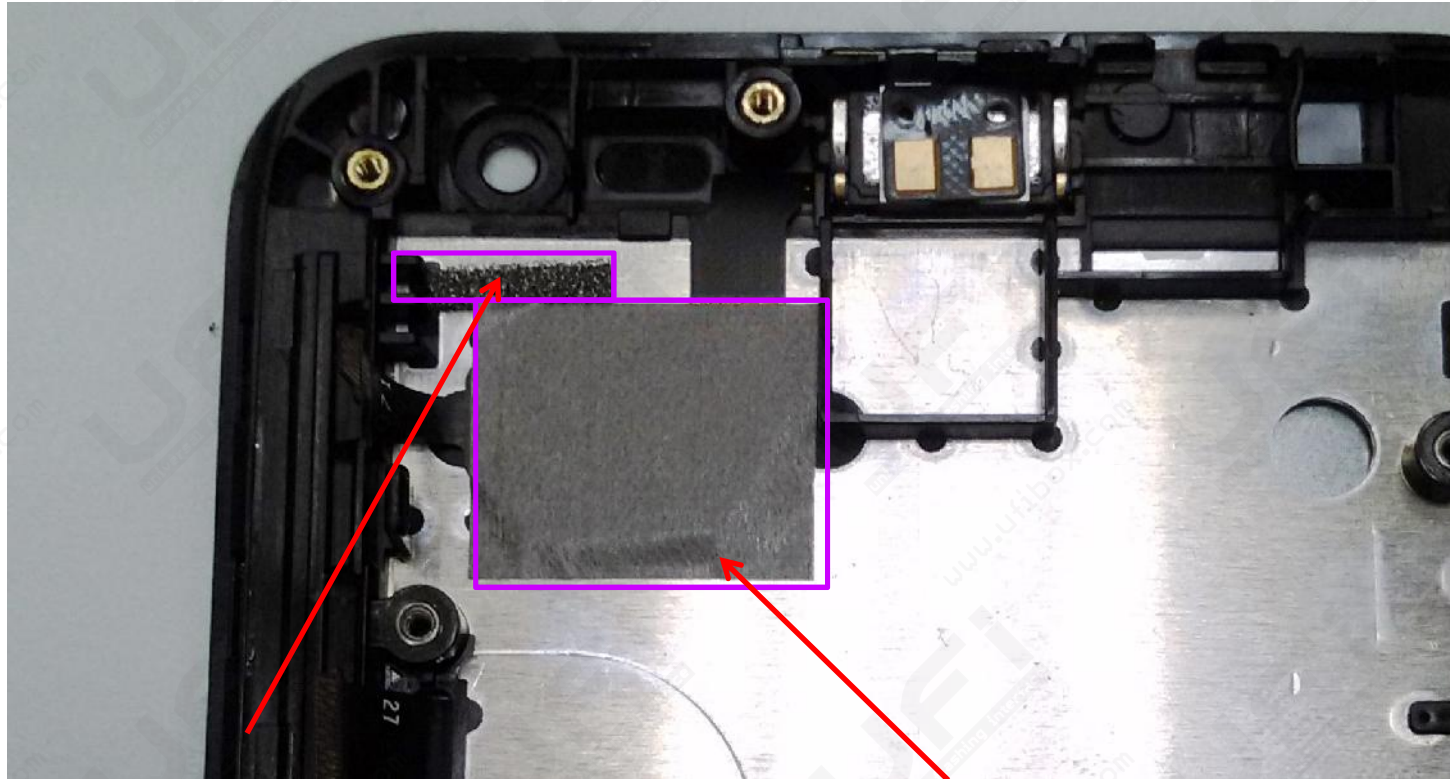
System level RF Test Trouble Shooting_2/3/4G (1/1)



TX power too low
Check gasket

**BER/RSRP/
RSCP/RSSI**
Check gasket

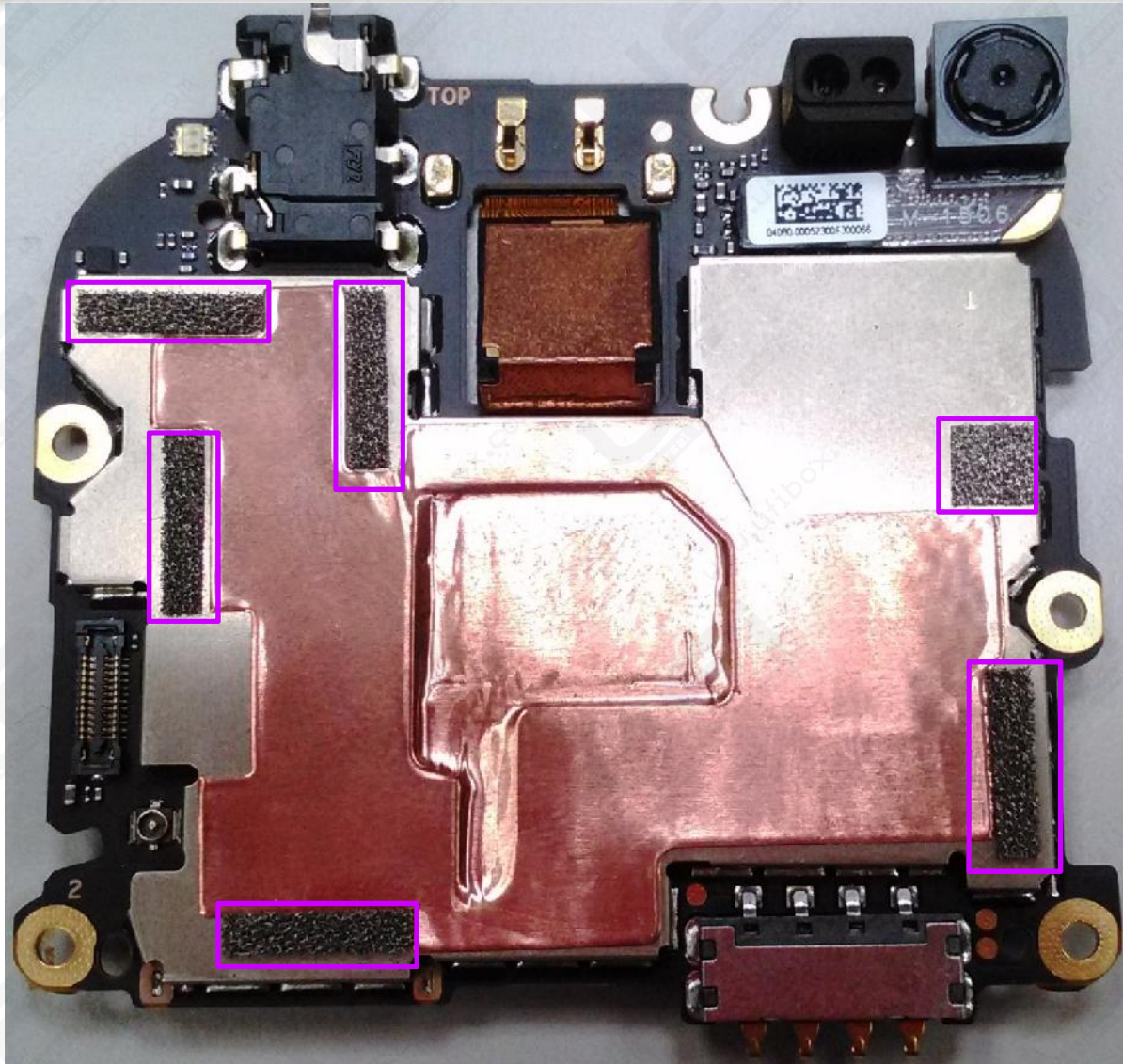
System level RF Test Trouble Shooting_ **WIFI/GPS** (1/4)



RX fail /GPS CN too low
Check gasket

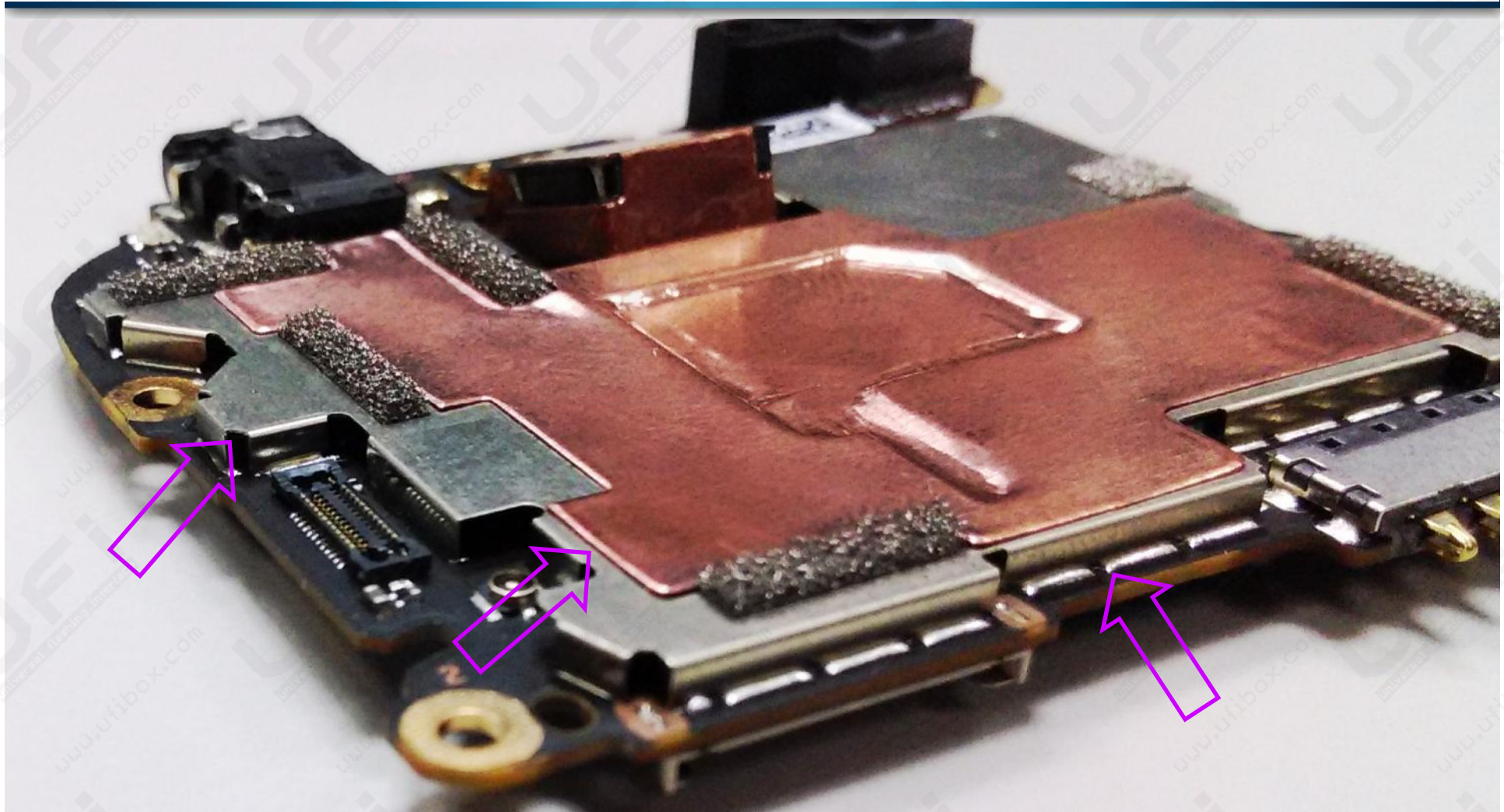
RX fail /GPS CN too low
Check the gasket location

System level RF Test Trouble Shooting_ **WIFI/GPS** (2/4)



All RX test item
include BT/WIFI/GPS

System level RF Test Trouble Shooting_WIFI/GPS(3/4)



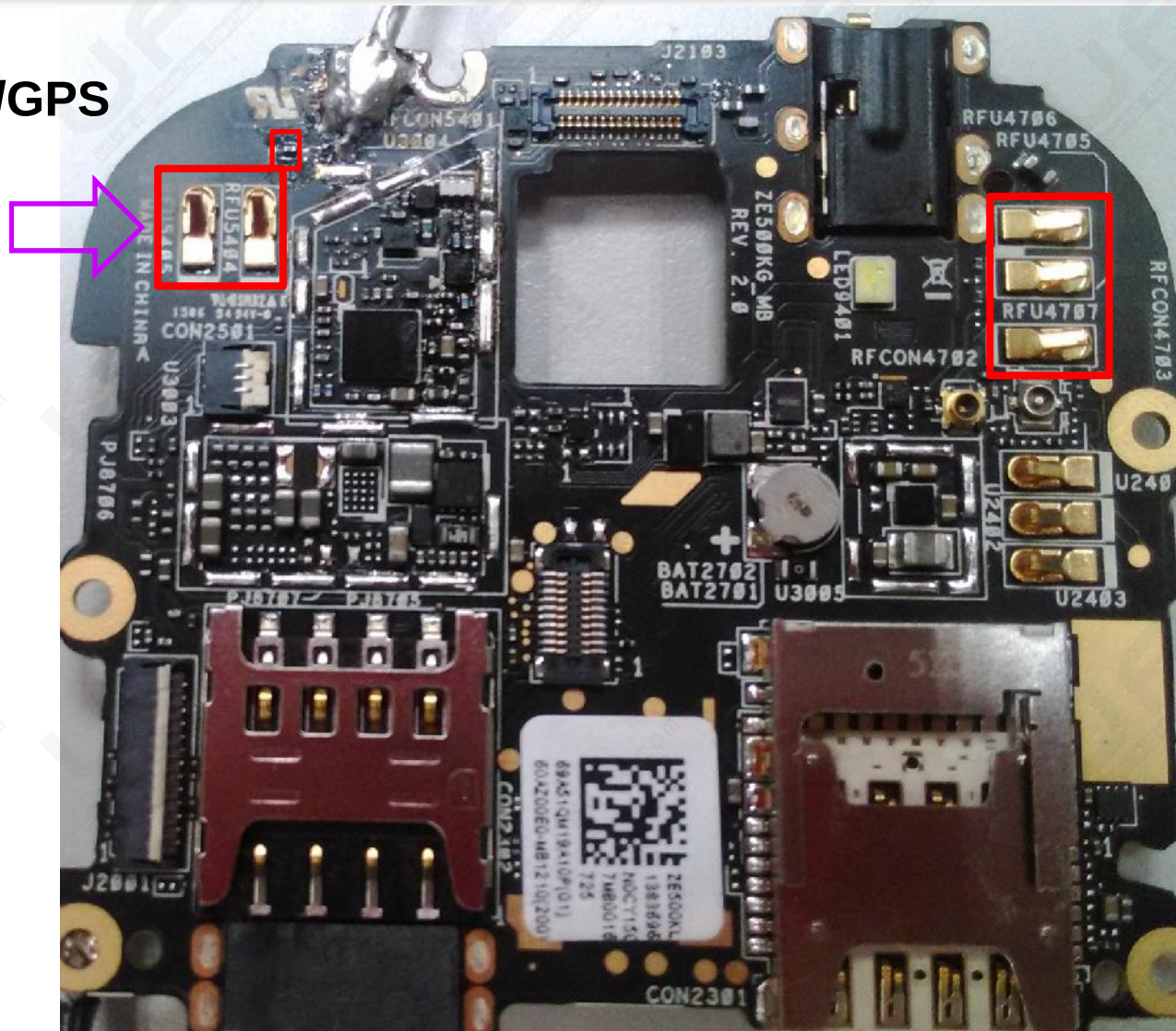
All RX test item include BT/WIFI/GPS

Check the shielding case & gasket

System level RF Test Trouble Shooting_WIFI/GPS(4/4)

BT/WIFI/GPS

Power/CN



Diversity RX

If following parts are moved, to be re-done Calibration (PHONE)

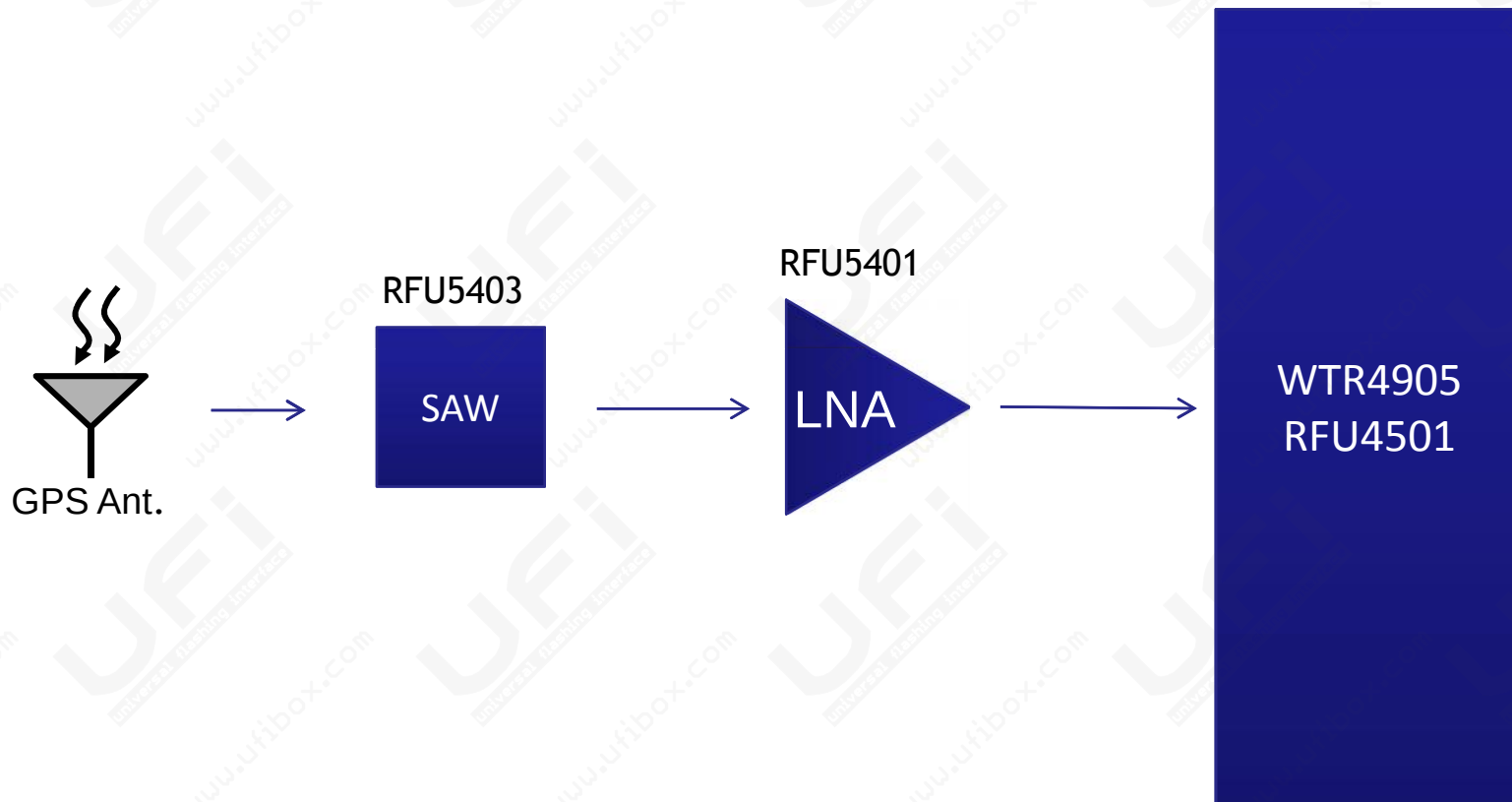
1. Transceiver: RFU4401 ASM: RFU4701/RFU4702
2. PA: RFU4813 /RFU5409 /RFU5001
3. Duplexer: RFU4801 /RFU4802 /RFU4805 /RFU4804
RFU4803 /RFU5009 /RFU5010 /RFU5002 /RFU5003
RFU5004
4. SAW: RFU5208 /RFU5201/ RFU5207/ RFU5206
RFU5213 /RFU5209 /RFU5210/ RFU5202
5. Coupler: RFCON4702
6. Switch: RFU5212/ RFU5008/ RFU5011/ RFU4809
RFU4704
7. Low Pass filter: RFU5006 /RFU4810 /RFU4502

Outline

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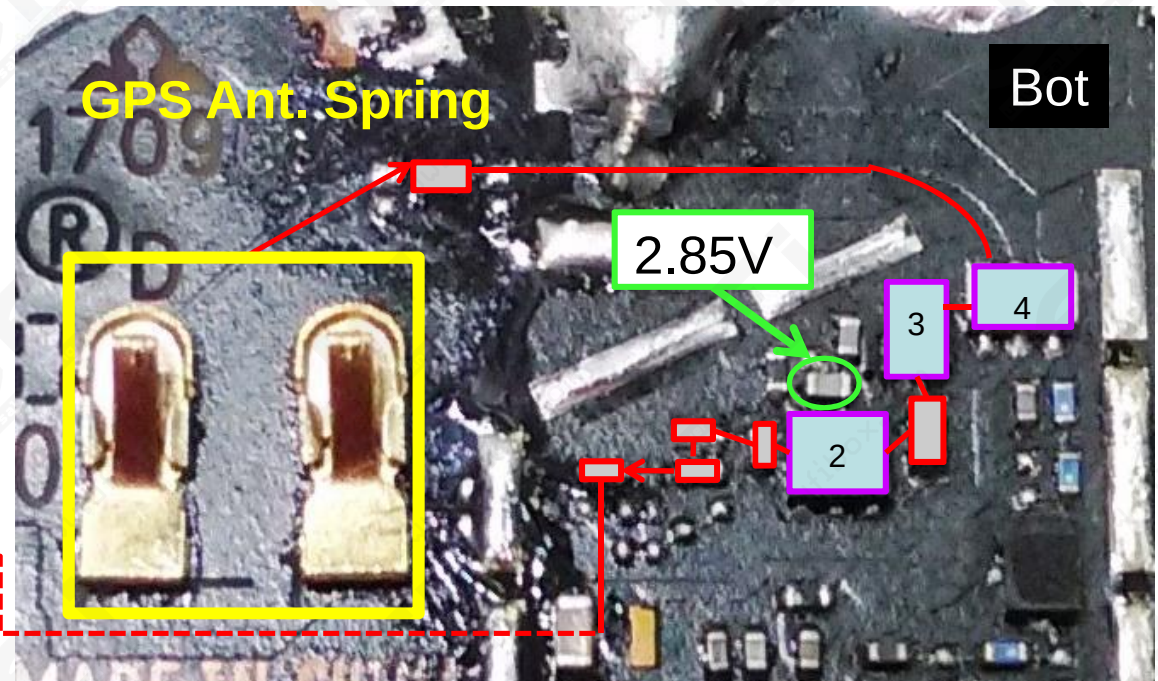
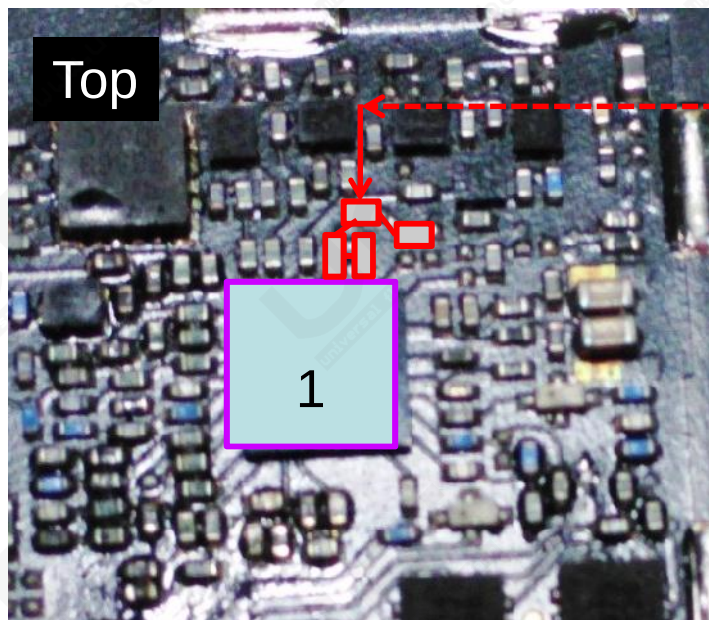
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GPS Ant Block Diagram



GPS

-  GPS Antenna area
-  Serial comp. with orientation
-  Serial comp. without orientation



1. RFU4501 : Transceiver
WTR4905
2. RFU5401: GPS LNA
3. RFU5403 : GPS SAW Filter
4. RFU5407: Diplexer

GPS Function Test Procedure

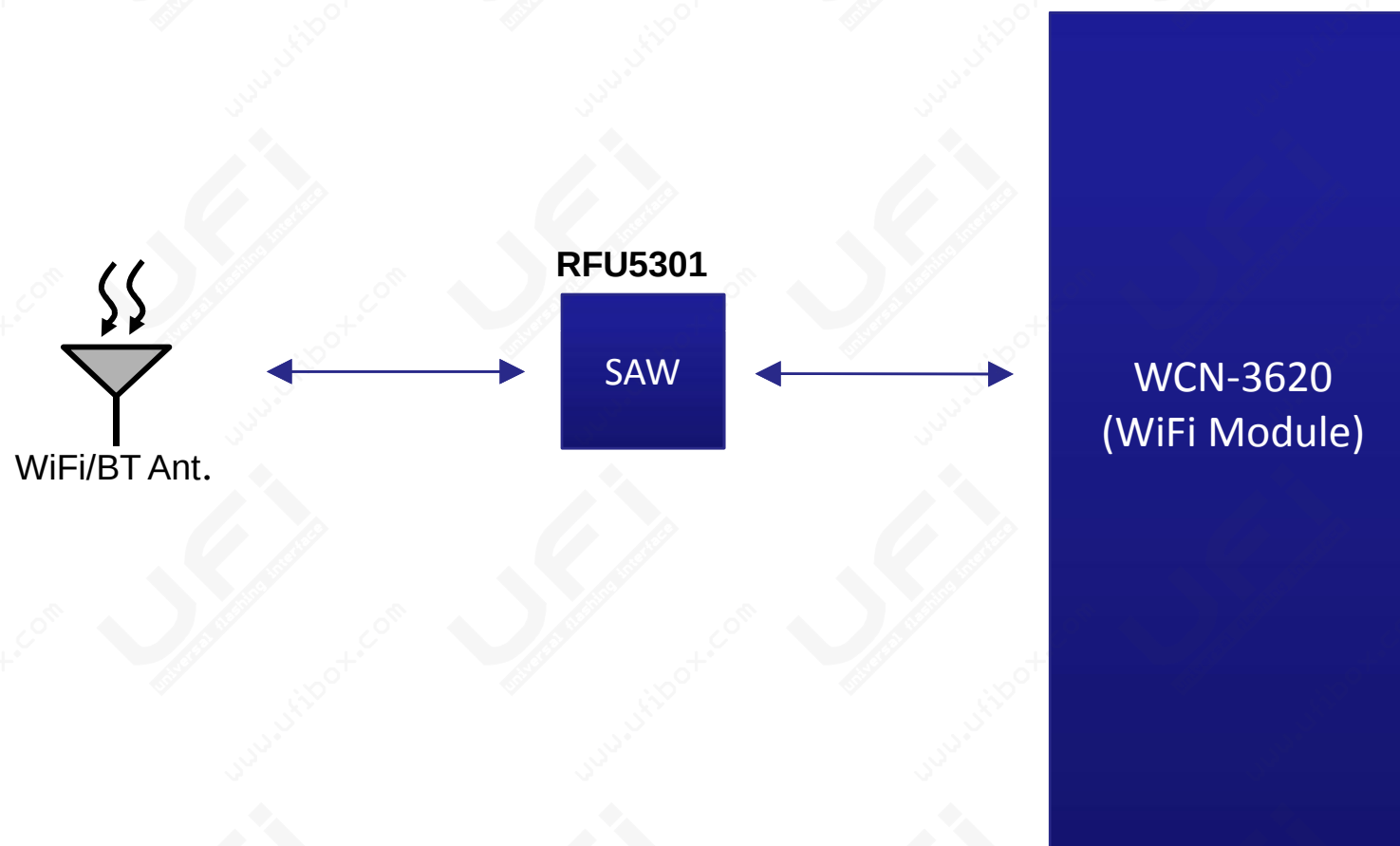
- **Check GPS function with GPS Test Tool.**
- **Check the GPS feed point.**
- **Check RFU5403 components**
- **Check the 2.85V supply voltage of GPS LNA**

Outline

(*) Debug Tool Required

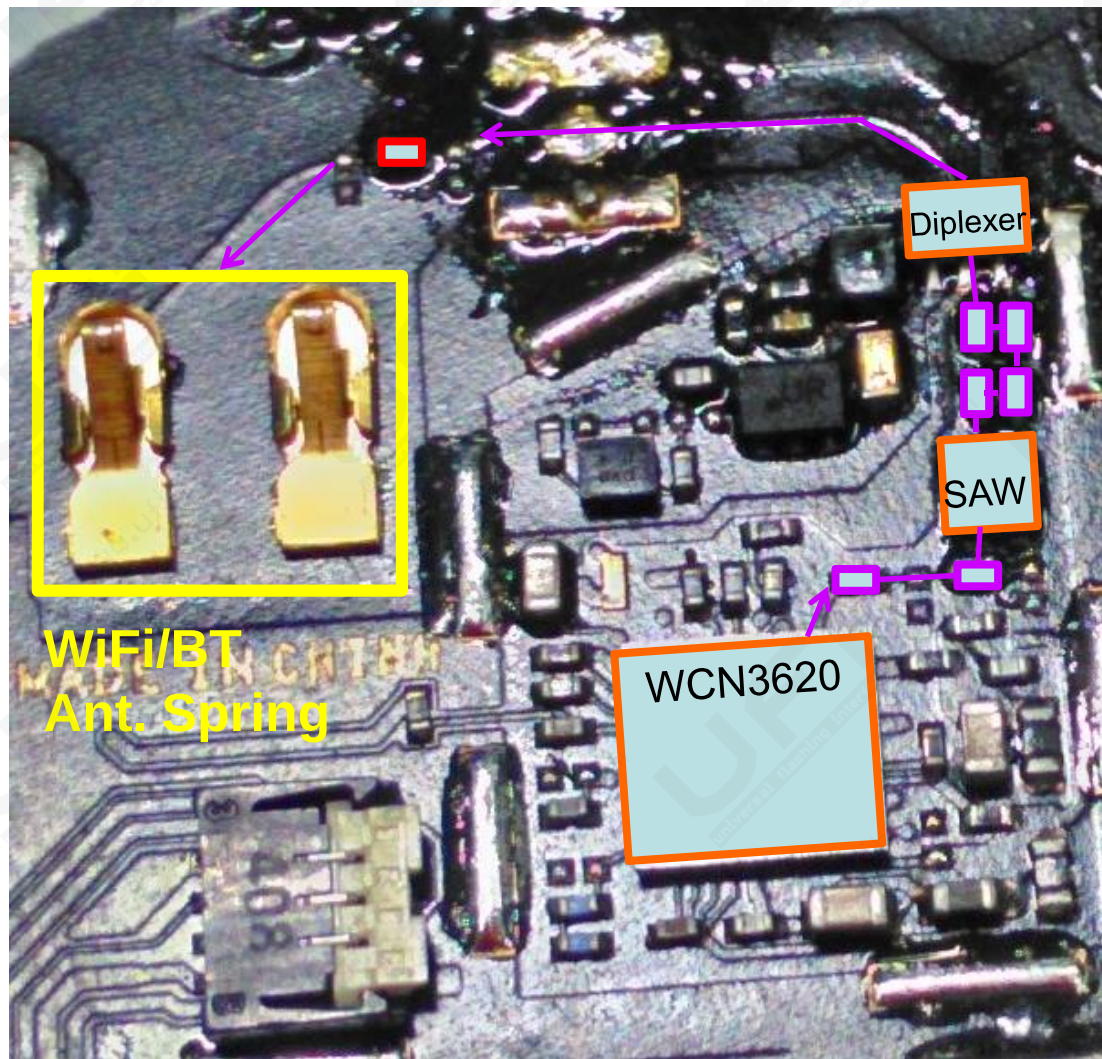
- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band8
 - LTE Quad-bands Block Diagram
 - LTE Band3/Band7/Band8/Band20/Band28
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band3/Band7/Band8/Band20/Band28
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFi Repair Guide

WiFi/BT Ant Block Diagram



WiFi/BT

-  WiFi Antenna area
-  Serial comp. without orientation
-  Serial comp. with orientation



BT/WLAN Function Test Procedure

Board level

- Check the BT/WLAN feed point
- Check WCN3620 and surrounding components

System level

- Change antenna
- Check the RFC5402

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